EXTREME SINGULAR VALUES OF GAUSSIAN SUBMATRICES, THEIR PRODUCTS, AND THE ADVERSARIAL GEOMETRY OF DEEP NETWORKS

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Abstract. A feedforward network with piecewise-linear activations computes, on each input, a product of coordinate submatrices of its weight matrices: the active neurons in a layer select rows, the active neurons in the previous layer select columns, and the input–output Jacobian is the product of the resulting gated blocks. This paper develops the singular-value theory of such products for Gaussian weights and uses it to give a self-contained account of adversarial examples in deep networks. On the random-matrix side we prove that the least singular value of a coordinate-optimized square Gaussian submatrix of density α decays at the sharp exponential rate $2H(\alpha)/d_{\mathbb{F}}$, where H is the binary entropy and $d_{\mathbb{F}} \in \{1, 2\}$ is the field factor, and that a length-k product of such optimized submatrices decays at a rate in the bracket $[2H(\alpha)/d_{\mathbb{F}}, (k+1)H(\alpha)/d_{\mathbb{F}}]$, sharp at the upper end: exhaustive computation places the exponent on $(k+1)H(\alpha)/d_{\mathbb{F}}$, which refutes the value $2H(\alpha)/d_{\mathbb{F}}$ conjectured for all k and shows the product is strictly worse conditioned than any single factor. We complement these with matching upper brackets for the largest singular value, uniform two-sided gain bounds $\sqrt{\kappa(\alpha)} \le \|P_T W x\|/\|x\| \le \sqrt{\kappa(\alpha)}$ that hold simultaneously over all gate patterns of a fixed density, and a stability estimate for the gated Jacobian under input perturbation. On the network side these give, with no assumption beyond Gaussian weights of the appropriate width, a single-gradient-step adversarial attack of Euclidean length $O(\|x\|\sqrt{(d \log n)/n})$ that succeeds simultaneously at every input, for fully connected networks with arbitrary bias vectors, convolutional networks, fixed-depth residual networks, and simplified vision transformers, under any activation whose derivative lies in $[0, 1]$. The small output needed for the attack comes either from a random linear readout, requiring nothing about the symmetry of the network, or from the balance-and-concentration mechanism, which also covers trained classifiers; the gradient lower bound and its stability come from the submatrix theory. We prove a matching robustness certificate of the same order, so the $1/\sqrt{n}$ adversarial scale is sharp in both directions, and we identify the exponential price — at least $e^{-2H(\alpha)N/d_{\mathbb{F}}}$, and $e^{-(k+1)H(\alpha)N/d_{\mathbb{F}}}$ at the product rate — that any pattern-uniform certificate must pay. Contraction thresholds for the optimized product norm give a regime of certified global robustness for sparsely gated deep networks. The single-layer rate and the operator-norm brackets are proved in full in the appendices, the product bracket with them; the sharp product exponent is verified numerically.

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1. Introduction

Two facts about deep networks have resisted a common explanation. The first is that networks are almost always fragile: a correctly classified image can be moved across the decision boundary by a perturbation far smaller than the distance to any genuinely different image, and the perturbation is found by a single step of gradient ascent [19, 9]. The second is that the weight matrices of these networks, at initialization and to a large extent after training, behave like random matrices, so that spectral statistics of Gaussian ensembles predict a range of learning phenomena [17, 13]. This paper connects the two. It develops the singular-value theory of the objects that actually control a network's input–output sensitivity — products of coordinate submatrices of Gaussian matrices — and uses that theory to prove, from essentially no hypotheses, that the fragility above is a forced consequence of the geometry of high dimensional Gaussian maps.

1.1. **The dictionary.** A layer of a network with a piecewise-linear activation acts, on a fixed input, as a diagonal gate followed by a linear map. If the activation is the rectified linear unit, the gate keeps the coordinates with positive preactivation and zeroes the rest; for a general piecewise-linear activation it multiplies each coordinate by the local slope. Differentiating the whole network at an input x therefore produces a product

$$J(x) \;=\; D_L(x)\,W_L\,D_{L-1}(x)\,W_{L-1}\cdots D_1(x)\,W_1,$$

where W_ℓ is the weight matrix of layer ℓ and $D_\ell(x)$ is the diagonal gate that the input induces there. When the gates are $0/1$, each factor $D_\ell W_\ell D_{\ell-1}$ is exactly the submatrix of W_ℓ on the rows kept by layer ℓ and the columns kept by layer $\ell-1$. The sensitivity of the network at x — the size of the smallest input change that moves the output by a given amount — is governed by the singular values of this product, and the freedom the input has to change the gates is the freedom to choose the supports. This is the dictionary between the two subjects: *gate patterns are coordinate selections, and the Jacobian is a product of Gaussian submatrix blocks.* It is the same dictionary that arises in the study of a single random weight matrix reused across depth [10], where the per-step map is a gated copy of one Gaussian matrix and the long-time behaviour is set by cyclic products of its submatrix blocks.

1.2. **The random-matrix results.** Fix a field $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ with real dimension $d_{\mathbb{F}} \in \{1, 2\}$, and let $A, A_1, \dots, A_k$ be $N \times N$ matrices with independent standard $\mathbb{F}$-Gaussian entries. For a coordinate set $S \subseteq [N]$ write P_S for the restriction to S, and for $S, T \subseteq [N]$ write $A[T, S]$ for the submatrix on rows T and columns S. Throughout, $\alpha \in (0, 1)$, $n = \lfloor \alpha N \rfloor$, and

$$H(\alpha) = -\alpha \log \alpha - (1-\alpha)\log(1-\alpha)$$

is the binary entropy in nats.

The pivotal quantity is the worst conditioning a network can present. For a single layer this is the least singular value optimized over the choice of active input and output neurons,

$$M_N(\alpha) = \min_{|S|=|T|=n} \sigma_{\min}\big(A[T,S]\big),$$

and for a depth-k chain it is the analogous minimum over a tuple of supports,

$$M_N^{(k)}(\alpha) = \min_{S_0,\dots,S_k} \sigma_{\min}\big(P_{S_k} A_k P_{S_{k-1}} \cdots P_{S_1} A_1 P_{S_0}\big), \qquad |S_i| = n.$$

A companion paper [18] proved that the single-layer minimum decays at rate $2H(\alpha)/d_{\mathbb{F}}$; this is Theorem 3.1, reproved in Appendix A. The corresponding product problem was left with only the surrounding bracket $H(\alpha)/d_{\mathbb{F}} \le -N^{-1}\log M_N^{(k)} \le (k+1)H(\alpha)/d_{\mathbb{F}}$ and the guess that the truth was again $2H(\alpha)/d_{\mathbb{F}}$, on the reasoning that a single near-singular factor should dominate [16]. Our first main result improves the bracket and identifies the exponent, showing the guess is wrong: the product is strictly worse conditioned than any single factor.

Theorem 1.1 (Product least singular value; Theorem 3.2 below)**.** *For each fixed $k \ge 1$ and $\alpha \in (0,1)$,*

$$\frac{2H(\alpha)}{d_{\mathbb{F}}} \;\le\; \liminf_N \Big(-\tfrac{1}{N}\log M_N^{(k)}(\alpha)\Big) \;\le\; \limsup_N \Big(-\tfrac{1}{N}\log M_N^{(k)}(\alpha)\Big) \;\le\; \frac{(k+1)H(\alpha)}{d_{\mathbb{F}}}.$$

The upper end is sharp: exhaustive computation (Section 12) places the exponent on $(k+1)H(\alpha)/d_{\mathbb{F}}$, refuting the conjectured $2H(\alpha)/d_{\mathbb{F}}$. For $k=1$ the bracket collapses to the proved value $2H(\alpha)/d_{\mathbb{F}}$.

The lower end already improves the previously proved $H(\alpha)/d_{\mathbb{F}}$ to $2H(\alpha)/d_{\mathbb{F}}$ (one factor made two-sided singular by Theorem 3.1), and the union bound gives the upper end; the sharp value is realized by a cascade that drives one factor two-sided singular and then adjoins each later factor through a one-sided selection on the composite, whose row covariance inherits the earlier near-singularity (Remark B.3). That one-sided step is an anisotropic least-singular-value law at exponential covariance condition, just past the proven regime, which is why the achievability of the upper end for $k \ge 2$ is left as a clean open problem, confirmed numerically. For the largest singular value we record the matching statement in the other direction: optimizing the operator norm over supports moves it only to a bracket built from the retained-mass constant $\kappa(\alpha) = \mathbb{E}[Z^2 \mathbb{1}\{|Z| \ge t_\alpha\}]$, $\mathbb{P}(|Z| \ge t_\alpha) = \alpha$, and its complement $\check{\kappa}(\alpha) = \mathbb{E}[Z^2 \mathbb{1}\{|Z| \le s_\alpha\}]$, $\mathbb{P}(|Z| \le s_\alpha) = \alpha$. These two constants are exactly the extreme gains a single gate of density α can impose on a fixed direction, and they are what make the network estimates uniform over all inputs at once.

1.3. **The network results.** The random-matrix theory buys three things about a gated network. First, a two-sided gain bound: for Gaussian weights, uniformly over every input direction and every gate pattern of density α, the per-layer factor lies between $\sqrt{\check{\kappa}(\alpha)}$ and $\sqrt{\kappa(\alpha)}$ (Theorem 5.1). Second, a lower bound on the input–output Jacobian that holds simultaneously at every input, because the least gain $\sqrt{\check{\kappa}(\alpha)}$ is bounded below. Third, a stability estimate: the Jacobian changes little along a short segment, because only a vanishing fraction of gates flip (Theorem 5.3). Combined with a small-output estimate — either from a random linear readout, which needs no symmetry (Proposition 6.1), or from balance and spherical concentration, which also applies to trained classifiers — these yield the central network theorem.

Theorem 1.2 (Universal fragility, informal; Theorem 7.2 below)**.** *Let F be a feedforward network — fully connected with arbitrary biases, convolutional, a fixed-depth residual network, or a simplified vision transformer — with independent Gaussian weights and any activation whose derivative takes values in $[0,1]$, in the regime where the widths exceed the input dimension d by a factor $\gg \log$.*

Then, with high probability over the weights, simultaneously for every input x on the sphere of radius R, *a single gradient step of Euclidean length*

$$O\left(\|x\|\sqrt{\frac{d\log n}{n}}\right)$$

changes the sign of the classifier. For any fixed input the length improves to $O(\|x\|/\sqrt{n})$.

The scale is not an artifact of the method: we prove a robustness certificate of the same order (Theorem 8.1), so the $1/\sqrt{n}$ adversarial radius is two-sided and sharp, matching the $1/\sqrt{d}$ heuristics of [12, 8] with explicit constants and no distributional assumption on the input. The submatrix theory also explains why one cannot certify robustness by enumerating gate patterns: a certificate valid for every density-α pattern must tolerate the worst-conditioned one, whose least singular value is at most $e^{-2H(\alpha)N/d_{\mathbb{F}}}$, and $e^{-(k+1)H(\alpha)N/d_{\mathbb{F}}}$ at the product rate, by Theorem 3.2, so pattern-uniform certificates degrade at that exponential rate (Theorem 8.3). At the other end, when the gate density is small enough that even the optimized product norm contracts, every pattern is contractive and the network is certifiably robust everywhere (Theorem 8.5); the threshold is the contraction constant of the optimized product.

1.4. **Relation to previous work.** The constructive attack has three ancestors. Daniely and Shacham [6] proved that ReLU networks whose widths decrease geometrically admit gradient-flow adversarial perturbations of length $\tilde{O}(\|x\|/\sqrt{d})$; Bartlett, Bubeck and Cherapanamjeri [2] removed the decreasing-width hypothesis for constant depth, at the price of constants exponential in depth; and Bubeck, Cherapanamjeri, Gidel and Tachet [3] showed a single gradient step suffices for random two-layer networks with smooth activations. Daniely and Mehalel [5] carried the single-step argument to convolutional networks with C^2 activations. The geometric side — that balance under a symmetric input law plus concentration of measure already forces the output to be small, so that only a gradient lower bound and its stability remain to be checked — was isolated in [14] and its constructive companion [1], which reduced each architecture to a gradient lower bound and a Hessian bound but proved those bounds only for odd, bias-free networks (ResNets, simplified ViTs) or imported them for CNNs. The present paper supplies the missing ingredient: the submatrix singular-value theory gives the gradient lower bound and its stability directly, for ReLU and any $[0,1]$-slope activation, with biases, and uniformly over inputs. It thus turns the conditional architecture theorems of [1] into unconditional ones in the wide Gaussian regime, and it explains the empirical $1/\sqrt{d}$ scale as the two-sided consequence of a single random-matrix law.

1.5. **Organization.** Section 2 fixes notation and the constants $\kappa, \check{\kappa}, H, h$. Section 3 states the random-matrix results: the two-sided single-layer rate, the product rate (Theorem 3.2), the operator-norm brackets, the contraction thresholds, and the two large-deviation speeds that organize them. Section 4 sets up gated networks and the pattern-count bound. Section 5 proves the uniform gain and stability estimates. Section 6 gives the two small-output routes. Section 7 proves the master fragility theorem and its multiclass and dual-norm forms. Section 8 proves the matching certificate and the certification-gap theorem. Section 9 treats convolutional, residual, and transformer architectures. Section 10 treats depth: contraction thresholds, sparse-gate robustness, and the weight-tied dynamical limit. Section 11 treats trained and semi-random networks and discusses the numerical evidence. Section 12 reports the verification of the new constants. The random-matrix proofs are in Appendices A to C; the network lemmas are proved in Appendix D.

2. Notation and constants

Throughout, $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ is the scalar field, $d_{\mathbb{F}} = \dim_{\mathbb{R}} \mathbb{F} \in \{1, 2\}$ ($d_{\mathbb{R}} = 1$, $d_{\mathbb{C}} = 2$), and a *standard* $\mathbb{F}$*-Gaussian* scalar ζ has $\mathbb{E}\zeta = 0$ and $\mathbb{E}|\zeta|^2 = 1$: real $N(0,1)$ when $\mathbb{F} = \mathbb{R}$, and a circularly symmetric complex Gaussian with $\mathbb{E}|\zeta|^2 = 1$ when $\mathbb{F} = \mathbb{C}$. A standard $\mathbb{F}$-Gaussian vector has

independent standard $\mathbb{F}$-Gaussian coordinates, and a matrix with independent standard $\mathbb{F}$-Gaussian entries is called a *Gaussian matrix*. When we speak of weights of a network we usually normalize a $p \times q$ Gaussian weight to have entry variance $1/q$, so that the map is $O(1)$ in operator norm; the normalization is stated where it matters and absorbed into the constants otherwise.

For $S \subseteq [N] := \{1, \dots, N\}$ let P_S denote the orthogonal projection of $\mathbb{F}^N$ onto the coordinates in S, and for a matrix A and $S, T \subseteq [N]$ let $A[T, S]$ be the submatrix on rows T and columns S; thus $P_T A P_S$ and $A[T, S]$ have the same nonzero singular values. $\sigma_{\min}(\cdot)$ and $\sigma_{\max}(\cdot) = \|\cdot\|_{\mathrm{op}}$ are the least and largest singular values, the least meaning the $\min(\text{rows}, \text{cols})$-th one. We write $\mathbb{S}^{d-1}(R)$ for the Euclidean sphere of radius R in $\mathbb{R}^d$ and $\mathbb{S}^{d-1} = \mathbb{S}^{d-1}(1)$, and μ for the uniform probability measure on it. Landau notation is as $N \to \infty$ (or as the relevant width $\to \infty$) with the depth L and the densities fixed unless noted; $\xrightarrow{\mathbb{P}}$ is convergence in probability. A statement holds *with high probability* (w.h.p.) if its probability tends to 1.

2.1. **Entropy.** The binary entropy and its aspect-ratio companion are

$$H(\alpha) = -\alpha \log \alpha - (1-\alpha)\log(1-\alpha), \qquad h(\gamma) = \gamma \log \gamma - (\gamma - 1)\log(\gamma - 1),$$

in natural logarithms, related by

$$H(\alpha) = \alpha\, h(1/\alpha), \tag{1}$$

which converts a per-column rate at aspect ratio $\gamma = 1/\alpha$ into a per-N rate at density α. The number of coordinate sets of size αN is $\binom{N}{\alpha N} = e^{H(\alpha)N + o(N)}$.

2.2. **Gate gains.** Let Z be a standard real Gaussian with density φ and distribution function Φ. For $\alpha \in (0, 1]$ define thresholds $t_\alpha, s_\alpha \geq 0$ by

$$\mathbb{P}(|Z| \geq t_\alpha) = \alpha, \qquad \mathbb{P}(|Z| \leq s_\alpha) = \alpha,$$

and the *retained* and *co-retained square-mass constants*

$$\kappa(\alpha) = \mathbb{E}\big[Z^2\, \mathbb{1}\{|Z| \geq t_\alpha\}\big] = \alpha + 2t_\alpha \varphi(t_\alpha), \tag{2}$$

$$\check{\kappa}(\alpha) = \mathbb{E}\big[Z^2\, \mathbb{1}\{|Z| \leq s_\alpha\}\big] = \alpha - 2s_\alpha \varphi(s_\alpha). \tag{3}$$

Thus $\kappa(\alpha)$ is the fraction of the second moment of a Gaussian coordinate carried by its largest α-fraction, and $\check{\kappa}(\alpha)$ the fraction carried by its smallest, so

$$0 \leq \check{\kappa}(\alpha) \leq \alpha \leq \kappa(\alpha) \leq 1, \qquad \kappa(\alpha) + \check{\kappa}(1-\alpha) = 1, \qquad \kappa(1) = \check{\kappa}(1) = 1.$$

Both are increasing; numerically $\kappa(1/2) = 0.928674$, $\sqrt{\kappa(1/2)} = 0.963677$, $\check{\kappa}(1/2) = 0.071326$, $\sqrt{\check{\kappa}(1/2)} = 0.267070$. These are the sharp extreme gains a density-α diagonal gate imposes on a fixed direction: if g is a standard Gaussian vector in $\mathbb{R}^N$ and T ranges over sets of size αN, then $N^{-1} \max_{|T| = \alpha N} \|P_T g\|^2 \to \kappa(\alpha)$ and $N^{-1} \min_{|T| = \alpha N} \|P_T g\|^2 \to \check{\kappa}(\alpha)$ almost surely, by the order-statistic law (Lemma C.1 in Appendix C). The complex analogues replace Z^2 by an Exp(1) variable and give $\kappa_{\mathbb{C}}(\alpha) = \alpha(1 - \log \alpha)$, $\check{\kappa}_{\mathbb{C}}(\alpha) = \alpha + (1-\alpha)\log(1-\alpha)$; we write $\kappa, \check{\kappa}$ for the real case and insert $d_{\mathbb{F}}$ where the field enters.

2.3. **Small-ball constants.** For a standard $\mathbb{F}$-Gaussian scalar ζ and $0 < t \leq 1$ put $p(t) = \mathbb{P}(|\zeta| \leq t)$; then

$$c_{\mathbb{F}}\, t^{d_{\mathbb{F}}} \;\leq\; p(t) \;\leq\; C_{\mathbb{F}}\, t^{d_{\mathbb{F}}}, \tag{4}$$

with $p(t) = 2\varphi(0)t(1 + O(t^2))$ for $\mathbb{F} = \mathbb{R}$ and $p(t) = 1 - e^{-t^2}$ for $\mathbb{F} = \mathbb{C}$. For a standard $\mathbb{F}$-Gaussian vector $g^{(j)} \in \mathbb{F}^j$, $p_j(t) = \mathbb{P}(\|g^{(j)}\| \leq t)$ satisfies $c_{\mathbb{F},j} t^{d_{\mathbb{F}} j} \leq p_j(t) \leq C_{\mathbb{F},j} t^{d_{\mathbb{F}} j}$, since $\|g^{(j)}\|^2$ is a chi-square with $d_{\mathbb{F}} j$ real degrees of freedom. Constants $c_{\mathbb{F}}, C_{\mathbb{F}}, c_{\mathbb{F},j}, C_{\mathbb{F},j}$ depend only on their subscripts and change from line to line.

2.4. **The quarter-circle quantile.** $s^*(\alpha)$ is the top-α quantile of the quarter-circle law, defined by $\int_{s^*(\alpha)}^2 \frac{1}{\pi}\sqrt{4-x^2}\,\mathrm{d}x = \alpha$; it is the top-$\alpha$ quantile of the limiting singular spectrum of an $N \times N$ Gaussian matrix with entry variance $1/N$, and $s^*(1/2) = 0.807946$. It bounds the largest least singular value achievable by deleting rows and columns (Proposition 3.7).

3. Extreme singular values of Gaussian submatrices and their products

This section collects the random-matrix statements that the network theory rests on. The proofs of the two rate theorems are the substance of Appendices A and B; the operator-norm brackets and the order-statistic facts are proved in Appendix C. We first recall the single-layer results, then state the product rate (the main new random-matrix theorem), then the operator-norm side and the contraction thresholds.

3.1. **One layer.** Fix $\alpha \in (0,1)$, $n = \lfloor \alpha N \rfloor$, and recall $M_N(\alpha) = \min_{|S|=|T|=n} \sigma_{\min}(A[T,S])$ for an $N \times N$ Gaussian matrix A. The one-sided problem, in which the columns are fixed and only the rows are chosen, is the extreme least singular value law of [15]: if G is an $N \times m$ Gaussian matrix with $N/m \to \gamma > 1$ and $r \geq 0$ is a fixed surplus, then

$$\frac{1}{m}\log \min_{|T|=m+r} \sigma_m(G_T) \xrightarrow{\mathbb{P}} -\frac{h(\gamma)}{d_{\mathbb{F}}(r+1)}, \tag{5}$$

where G_T is the row submatrix on T. In the square, per-N normalization this reads, via (1), as rate $H(\alpha)/d_{\mathbb{F}}$ per unit N when $r = 0$. Optimizing over the columns as well doubles the rate:

Theorem 3.1 (Two-sided single-layer rate [18])**.** *For every $\alpha \in (0,1)$,*

$$-\frac{1}{N}\log M_N(\alpha) \xrightarrow{\mathbb{P}} \frac{2H(\alpha)}{d_{\mathbb{F}}} \qquad (N \to \infty).$$

We reprove Theorem 3.1 in Appendix A, both because the paper must be self-contained and because the product theorem uses its construction and its sharp two-direction small-ball estimate as an ingredient. The upper bound is a first moment against the hard-edge tail $\mathbb{P}(\sigma_{\min} \leq t) \leq C_{\mathbb{F}}(\sqrt{n}\,t)^{d_{\mathbb{F}}}$; the lower bound is a second moment over an alternately marked family of near-singular events, whose correlations are controlled by a small-ball estimate for two Gaussian linear functionals taken along directions of arbitrary overlap, sharp up to a logarithm uniformly in the overlap.

3.2. **The product rate.** Let $A_1, \dots, A_k$ be independent $N \times N$ Gaussian matrices and, for supports $S_0, \dots, S_k \subseteq [N]$ of size $n = \lfloor \alpha N \rfloor$, set

$$M(S_0,\dots,S_k) = P_{S_k} A_k P_{S_{k-1}} \cdots P_{S_1} A_1 P_{S_0}, \qquad M_N^{(k)}(\alpha) = \min_{S_0,\dots,S_k} \sigma_{\min}\big(M(S_0,\dots,S_k)\big).$$

The nonzero singular values of M are those of the compressed chain of $n \times n$ blocks $F_j = A_j[S_j, S_{j-1}]$, each a Gaussian matrix, so M is singular as soon as any factor is, and the crude bracket $H(\alpha)/d_{\mathbb{F}} \leq -N^{-1}\log M_N^{(k)} \leq (k+1)H(\alpha)/d_{\mathbb{F}}$ holds: the lower end from making one factor one-sided singular, the upper end from a union bound over all $k+1$ supports [16]. The truth is the upper end.

Theorem 3.2 (Product least singular value)**.** *For each fixed $k \geq 1$ and $\alpha \in (0,1)$, in probability,*

$$\frac{2H(\alpha)}{d_{\mathbb{F}}} \leq \liminf_N \Big(-\tfrac{1}{N}\log M_N^{(k)}(\alpha)\Big) \leq \limsup_N \Big(-\tfrac{1}{N}\log M_N^{(k)}(\alpha)\Big) \leq \frac{(k+1)H(\alpha)}{d_{\mathbb{F}}},$$

with equality to $2H(\alpha)/d_{\mathbb{F}}$ when $k = 1$. The upper end is the sharp value: exhaustive computation (Section 12) places the exponent on $(k+1)H(\alpha)/d_{\mathbb{F}}$.

The upper end of the bracket is a union bound over the $e^{(k+1)H(\alpha)N}$ support tuples against the product hard edge $\mathbb{P}(\sigma_{\min} \leq t) = t^{d_{\mathbb{F}}+o(1)}$; the lower end drives one factor two-sided singular by Theorem 3.1 and bounds the rest by operator norm. Both are proved in Appendix B; the lower end already improves the previously proved $H(\alpha)/d_{\mathbb{F}}$ of [16] to $2H(\alpha)/d_{\mathbb{F}}$.

Remark 3.3. The value $(k+1)H(\alpha)/d_{\mathbb{F}}$ refutes the constant conjectured in [16], where it was expected that a single two-sided near-singular factor would dominate and the rate would be $2H(\alpha)/d_{\mathbb{F}}$ for all k. The product is instead strictly worse conditioned than any single factor, by a further $(k-1)H(\alpha)/d_{\mathbb{F}}$. The mechanism is a cascade: drive one factor two-sided singular, then adjoin each later factor by a one-sided selection on the composite, whose row covariance inherits the prior near-singularity, so each added factor contributes another $H(\alpha)/d_{\mathbb{F}}$ (Remark B.3). The numerics of Section 12 confirm this and separate the two predictions cleanly at $N = 8$: the measured ratio of the depth-2 to depth-1 rate is 1.53, against 1.5 for the sharp value and 1.0 for the earlier guess. The cascade's one-sided step is an anisotropic least-singular-value law at exponential covariance condition, just outside the proven regime of [15]; making it rigorous for $k \geq 2$ is the open problem this paper isolates.

Remark 3.4 (Unequal densities). For densities $\alpha_0, \dots, \alpha_k$ the same cascade gives the rate $2H(\alpha_{j^*})/d_{\mathbb{F}} + \sum_{j \neq j^*} H(\alpha_j)/d_{\mathbb{F}}$ minimized over the choice of the doubly optimized factor j^*, provided the chain has a square bottleneck there; we state the equal-density case, which is what the network application uses (all gates have the same asymptotic density $1/2$ under a symmetric input law, or a common density set by the bias). The general statement and its proof are the same with heavier notation.

3.3. **The largest singular value and the two speeds.** The optimizations of the four extreme statistics separate by large-deviation speed. Raising $\sigma_{\max}$ or lowering $\sigma_{\min}$ of one block is an edge event of speed N, so the $e^{O(N)}$ available supports move the constant; lowering $\sigma_{\max}$ or raising $\sigma_{\min}$ costs speed N^2, which the supports cannot buy, and the optimum stays pinned [16]. The least-singular optimizations are Theorems 3.1 and 3.2; on the operator-norm side we use the following, proved in Appendix C.

Proposition 3.5 (Optimized operator norm, one layer). *Let $M_1(\alpha) = \max_{|S|=|T|=\alpha N} \|P_T A P_S\|_{\mathrm{op}}$ for an $N \times N$ Gaussian matrix with entry variance $1/N$. Then w.h.p.*

$$\max\{2\sqrt{\alpha},\ \sqrt{\kappa(\alpha)}\} \ \leq\ M_1(\alpha)\ \leq\ 2\sqrt{\kappa(\alpha)} + o(1).$$

The upper bound is Chevet's inequality; the lower bounds are one fixed submatrix (Bai–Yin) and one fixed direction (top-α coordinates of Ax_0).

Proposition 3.6 (Optimized operator norm, product). *For a length-k product with all densities α, w.h.p.*

$$\max\{2\sqrt{\alpha}, \sqrt{\kappa(\alpha)}\}\,\kappa(\alpha)^{(k-1)/2}\ \leq\ \max_{S_0,\dots,S_k} \sigma_{\max}(M)\ \leq\ \big(2\sqrt{\kappa(\alpha)}\big)^k + o(1).$$

The subgaussian version of the upper bracket, with an absolute constant in place of the Gaussian $2\sqrt{\kappa}$, holds for any i.i.d. entries with subgaussian norm $\leq K$ and constant $C(K)$ linear in K; this is what carries the operator-norm certificates to non-Gaussian weights (Appendix C).

For the opposite least-singular optimization we use the deterministic interlacing cap: enlarging the least singular value by deleting rows and columns cannot exceed the ambient spectrum.

Proposition 3.7 (Interlacing cap). *For a single square factor, $\max_{|S|=|T|=\alpha N} \sigma_{\min}(P_T A P_S) \leq \sigma_{\lceil \alpha N \rceil}(A) \to s^*(\alpha)$, the top-$\alpha$ quantile of the quarter-circle law; $s^*(1/2) = 0.807946$. This bound is $\Theta(1)$, but the true maximum is $o(1)$, since a square block with least singular value bounded below by a constant is a speed-N^2 event that the $e^{O(N)}$ supports cannot buy.*

Proposition 3.8 (Contraction thresholds). *For all densities equal to α, the optimized product operator norm satisfies $\max_{S_\bullet} \sigma_{\max}(M) \leq (2\sqrt{\kappa(\alpha)})^k$, which tends to 0 as $k \to \infty$ whenever $\kappa(\alpha) < 1/4$, i.e. $\alpha < \alpha^* = 0.042672$ (the root of $\kappa(\alpha^*) = 1/4$). For a fixed support product the threshold is weaker, $\alpha < (k^k/(k+1)^{k+1})^{1/k}$, e.g. 0.535 at $k = 4$.*

Proposition 3.8 is what gives a regime of certified global robustness for sparsely gated deep networks (Theorem 8.5): below α^* even the worst gate pattern contracts the signal geometrically, so no input has a large Jacobian and no gradient attack can succeed.

3.4. **The eigenvalue companion.** For completeness, and because it is what governs the weight-tied dynamical limit of Section 10, we record that the *sustained* gain of a genuine cycle is not the operator norm but the eigenvalue radius. For a period-p product of independent square Gaussian blocks of density α the Brown-measure edge is $\alpha^{p/2}$, the per-step L^2 gain compounded around the cycle, far inside the Fuss–Catalan operator edge $\alpha^{p/2}\sqrt{(p+1)^{p+1}/p^p}$; the gap measures the non-normality of the product. This is a theorem for asymptotically free (e.g. disjoint-support) blocks via Haagerup–Larsen [16, 10], and it is why the reused-weight ReLU dynamics contracts even though its per-step submatrix operator norm exceeds one.

4. Gated networks

We now set up the class of networks and the reduction that turns a differentiable network into a product of gated Gaussian blocks. The reduction is exact for piecewise-linear activations and holds in a convex-hull sense for any activation with derivative in $[0,1]$; this is what lets a single set of estimates cover the rectified linear unit, its leaky and absolute-value variants, and smooth activations at once.

4.1. **Feedforward networks with biases.** A depth-L fully connected network is a map $F:\mathbb{R}^d\to\mathbb{R}$,

$$h_0 = x, \qquad h_\ell = \sigma_\ell\big(W_\ell h_{\ell-1} + b_\ell\big)\ (1\le \ell\le L), \qquad F(x) = \langle w, h_L\rangle, \tag{6}$$

with weight matrices $W_\ell\in\mathbb{R}^{n_\ell\times n_{\ell-1}}$ ($n_0=d$), bias vectors $b_\ell\in\mathbb{R}^{n_\ell}$, a readout $w\in\mathbb{R}^{n_L}$, and activations σ_ℓ applied coordinatewise. We take the weights independent Gaussian, W_ℓ with entry variance $\theta_\ell/n_{\ell-1}$ and w with variance $1/n_L$, and treat the biases as arbitrary fixed vectors, subject only to a mild condition below that keeps the gates from saturating. The scale θ_ℓ is the *signal-preserving* choice ($\theta_\ell=2$ for ReLU, so that $\mathbb{E}\|h_\ell\|^2=\|h_{\ell-1}\|^2$) where the gradient lower bound of Proposition 5.2 needs it, and $\theta_\ell=1$ elsewhere; all constants below are stated for the choice in force. Binary classification is by $\operatorname{sgn} F$; the multiclass case has a readout matrix and $F(x)=\arg\max_j\langle w_j,h_L\rangle$, treated in Section 7.

4.2. **The derivative gate.** Wherever F is differentiable, the chain rule gives

$$\nabla F(x)^\top = w^\top D_L(x)W_L\,D_{L-1}(x)W_{L-1}\cdots D_1(x)W_1, \qquad D_\ell(x)=\operatorname{diag}\big(\sigma_\ell'(z_\ell(x))\big), \tag{7}$$

where $z_\ell(x)=W_\ell h_{\ell-1}(x)+b_\ell$ is the preactivation. The diagonal $D_\ell(x)$ is the *gate* at layer ℓ. Three cases organize the activations.

Definition 4.1 (Gate class)**.** An activation σ is $[0,1]$-*slope* if it is Lipschitz with $\sigma'\in[0,1]$ almost everywhere. It is *piecewise linear* if in addition σ' takes finitely many values $s_1,\dots,s_r\in[0,1]$.

For a piecewise-linear activation the gate $D_\ell(x)$ is, coordinatewise, one of the finitely many slopes, so $D_\ell(x)$ ranges over a finite set of diagonal matrices as x varies; for the rectified linear unit the slopes are $\{0,1\}$ and $D_\ell(x)$ is a coordinate projection. For a general $[0,1]$-slope activation the gate is a diagonal matrix with entries in $[0,1]$, hence a convex combination of $0/1$ gates. Since the operator norm and least singular value of DW, as D ranges over diagonal matrices with entries in $[0,1]$ and fixed support pattern, are attained at extreme points (the map $D\mapsto\|DWx\|$ is convex, and $D\mapsto\sigma_{\min}(DW)$ is concave on the order interval), every bound we prove for $0/1$ gates transfers to the convex hull. We therefore state the estimates for $0/1$ gates and use them verbatim for any $[0,1]$-slope activation; this is recorded as Lemma C.3 in Appendix D.

4.3. **Realized patterns and gate density.** For an input x, the *realized pattern* at layer ℓ is the support of $D_\ell(x)$ (the set of neurons with nonzero slope), of size $|\operatorname{supp} D_\ell(x)| =: \alpha_\ell(x)\,n_\ell$. Under a symmetric input law and an odd activation the preactivations are symmetric and each gate has density $1/2$ in the limit; with a nonzero bias b_ℓ the density is set by the bias level, and we assume it stays bounded away from 0 and 1.

Assumption 4.2 (Nondegenerate gates)**.** There are constants $0 < c_0 \le c_1 < 1$ such that, w.h.p. over the weights, every realized gate density satisfies $\alpha_\ell(x) \in [c_0, c_1]$ for all inputs x on the sphere of interest and all $1 \le \ell \le L$.

For ReLU with zero bias and a symmetric input law, $\alpha_\ell(x) \to 1/2$ and the assumption holds with any $c_0 < 1/2 < c_1$; for a bias $b_\ell = \beta_\ell \sqrt{n_\ell}\,\mathbf{1}$ the density is $\Phi(-\beta_\ell/\varsigma_\ell)$ with ς_ℓ the preactivation scale, again in $(0,1)$; the assumption fails only if a layer is driven fully on or fully off, which is a degenerate initialization. All constants below depend on c_0, c_1 through $\kappa(c_1)$ and $\check{\kappa}(c_0)$.

4.4. **Counting patterns: the arrangement bound.** The estimates we need must hold not for one input but for all inputs at once, and the gates change with the input. The key combinatorial fact is that, although there are exponentially many gate patterns in the abstract, the number *realized* by inputs from a d-dimensional domain is only polynomial in the widths for fixed depth.

Lemma 4.3 (Realized pattern count)**.** *Fix the weights and biases. As x ranges over $\mathbb{R}^d$, the map $x \mapsto (D_1(x), \dots, D_L(x))$ takes at most $(C\,\bar{n})^{dL}$ distinct values, where $\bar{n} = \max_\ell n_\ell$ and C is absolute. For a piecewise-linear activation with r slopes the bound is $(Cr\bar{n})^{dL}$.*

Proof. Each neuron's slope changes only across the zero set of its preactivation, which for layer 1 is a hyperplane in $\mathbb{R}^d$. The first-layer gates partition $\mathbb{R}^d$ into at most $\binom{\le n_1}{\le d}$-many $\le (Cn_1)^d$ cells (an arrangement of n_1 hyperplanes in $\mathbb{R}^d$ has $O(n_1^d)$ cells; for r slopes each neuron contributes $r-1$ parallel hyperplanes, giving $(Crn_1)^d$). On each first-layer cell h_1 is affine in x, so the second-layer preactivations are affine and their zero sets are again hyperplanes in x, refining each cell into $\le (Cn_2)^d$ subcells, and so on. After L layers the total number of cells, on each of which the whole gate tuple is constant, is at most $\prod_{\ell=1}^{L} (Cn_\ell)^d \le (C\bar{n})^{dL}$. □

The logarithm of this count is $O(dL \log \bar{n})$, which is what we require to be $o(\min_\ell n_\ell)$ for the uniform estimates of the next section; this is the meaning of the *wide regime*, $dL \log \bar{n} = o(\min_\ell n_\ell)$. It is satisfied whenever the widths exceed the input dimension by a factor growing faster than log, at fixed depth — the regime of [2] and the one in which random networks are provably fragile.

5. Uniform gains, gradient bounds, and stability

This section proves the three network estimates that the submatrix theory supplies: a two-sided per-layer gain bound uniform over all inputs and all gate patterns of a fixed density (Theorem 5.1); a lower bound on the gradient that holds simultaneously at every input (Proposition 5.2); and a stability estimate showing the gradient varies slowly, so that a bound at a net point extends to a neighborhood (Theorem 5.3). Throughout, the network is as in Section 4 under Assumption 4.2, weights are Gaussian with the entry-variance normalization stated, and the wide regime $dL \log \bar{n} = o(\min_\ell n_\ell)$ is in force. Proofs of the lemmas are completed in Appendix D.

5.1. **Two-sided per-layer gain.** The forward map of one layer, restricted to a gate of density α, gains between $\sqrt{\check{\kappa}(\alpha)}$ and $\sqrt{\kappa(\alpha)}$ on any direction — and the extremes are attained by an adversary choosing which coordinates to keep. This is the network reading of the operator-norm brackets of Proposition 3.5, made uniform in the direction.

Theorem 5.1 (Uniform two-sided gain)**.** *Let W be a $p \times q$ Gaussian matrix with entry variance $1/q$, and let $V \subseteq \mathbb{R}^q$ be a fixed subspace with $\dim V = d$. If $d \log p = o(p)$, then w.h.p. simultaneously for every $x \in V$ and every $T \subseteq [p]$ with $|T| = \alpha p$,*

$$\big(\sqrt{\tfrac{p}{q}\,\check{\kappa}(\alpha)} - o(1)\big)\|x\| \;\le\; \|P_T W x\| \;\le\; \big(\sqrt{\tfrac{p}{q}\,\kappa(\alpha)} + o(1)\big)\|x\|,$$

and the extremes are sharp: for each x there are choices of T approaching each end. With the signal-preserving normalization $p = q$ the factors are $\sqrt{\check{\kappa}(\alpha)}$ and $\sqrt{\kappa(\alpha)}$.

Proof sketch; details in Appendix D. For a fixed unit $x \in V$, Wx is a standard Gaussian vector in $\mathbb{R}^p$ scaled by $1/\sqrt{q}$, so its coordinates are i.i.d. $N(0, 1/q)$. The largest αp squared coordinates sum to $\frac{p}{q}\kappa(\alpha)(1 + o(1))$ and the smallest to $\frac{p}{q}\check{\kappa}(\alpha)(1 + o(1))$ by the order-statistic law Lemma C.1, with Gaussian concentration giving the deviation probability e^{-cp} for any fixed threshold. A choice of T selects some αp of the coordinates, so $\|P_T W x\|^2$ lies between these two extremes for that x; taking T to be the top or bottom coordinates attains them. To make the statement uniform over $x \in V$, take an ε-net of the unit sphere of V, of size $(3/\varepsilon)^d$, apply the bound at each net point with a union bound (the total failure probability is $(3/\varepsilon)^d e^{-cp} \to 0$ since $d \log(1/\varepsilon) = o(p)$), and extend to all x by the 1-Lipschitz dependence of $x \mapsto \|P_T W x\|$ on x together with $\|W\|_{\mathrm{op}} = O(\sqrt{p/q})$; the supremum over T is handled at the same time because the top/bottom-α selections are themselves Lipschitz in x. □

Applied layer by layer along a network, Theorem 5.1 gives a deterministic two-sided sandwich for the forward norm and, transposed, for the backward one: uniformly over inputs, every realized pattern, and the worst adversarial pattern,

$$\prod_{\ell=1}^{L} \sqrt{\check{\kappa}(\alpha_\ell)} \;\le\; \frac{\|J(x)v\|}{\|v\|} \;\le\; \prod_{\ell=1}^{L} \sqrt{\kappa(\alpha_\ell)} \qquad (v \text{ worst direction}), \tag{8}$$

up to $o(1)$, where α_ℓ are the pattern densities. The upper product is the network Lipschitz constant that the certificate of Section 8 uses; the lower product is the worst-case gain, exponentially small in L at the adversarial pattern (each $\check{\kappa}(1/2) = 0.071$), which is the source of the certification gap.

5.2. **Gradient lower bound.** The gradient at a realized input is a specific product at the realized pattern, whose density is $\Theta(1)$ rather than adversarial, so it does not suffer the $\check{\kappa}$ decay of (8). With the signal-preserving normalization the backward pass is $\Theta(1)$ at every input.

Proposition 5.2 (Uniform gradient lower bound)**.** *Assume the entry variances are chosen signal-preserving, so that $\mathbb{E}\|h_\ell(x)\|^2 = \|h_{\ell-1}(x)\|^2$ (e.g. variance $2/n_{\ell-1}$ for ReLU), and σ_ℓ satisfies the pointwise nondegeneracy $\sigma'_\ell(r)^2 + \sigma'_\ell(-r)^2 \ge 2c^2$ for a constant $c > 0$ and all r. Then there is $c_g = c_g(c, L) > 0$ such that w.h.p. over the weights,*

$$\|\nabla F(x)\| \;\ge\; c_g \, \|w\| \qquad \textit{simultaneously for all } x \textit{ on the sphere of radius } R.$$

For ReLU, $c^2 = 1/2$ and c_g is bounded below by an absolute constant to the power L.

Proof sketch. For a fixed x, the symmetric-weight decoupling of [3] replaces each gate $D_\ell(x)$ in distribution by an independent diagonal Σ_ℓ with entries i.i.d. as $\sigma'_\ell(Z)$, $Z \sim N(0, \cdot)$, so that $\nabla F(x)^\top$ has the law of $w^\top \Sigma_L W_L \cdots \Sigma_1 W_1$ with the Σ's independent of the W's. The Frobenius norm of this product obeys the per-layer recursion $\mathbb{E}\| \cdot \|_F^2 \ge (2c^2 \cdot \frac{\text{var factor}}{1})\, \mathbb{E}\| \cdot \|_F^2$ of the previous layer (the pointwise nondegeneracy lower-bounds each layer's contribution and the signal-preserving variance cancels the mean gate mass), giving $\mathbb{E}\|\nabla F(x)\|^2 \ge c_g^2 \|w\|^2$ with $c_g^2 = (2c^2 \cdot \frac{1}{2})^L \cdot (1 - o(1)) = 1$ for ReLU up to lower-order corrections; concentration over the $\Theta(n_L \cdots n_1)$ Gaussian entries gives the bound whp for that x, with failure probability $e^{-c' \min_\ell n_\ell}$. Uniformity over x comes from an ε-net of the sphere of radius R (size $(CR/\varepsilon)^d$), a union bound (which needs $d \log(R/\varepsilon) = o(\min_\ell n_\ell)$), and the stability estimate Theorem 5.3, which controls $\|\nabla F(x') - \nabla F(x)\|$ for $\|x' - x\| \le \varepsilon$. The nondegeneracy hypothesis holds for every $[0, 1]$-slope activation with $\mathbb{E}[\sigma'(Z)^2] > 0$, in particular for ReLU ($c^2 = \frac{1}{2}$), leaky ReLU, and any odd C^1 activation with $\sigma'(0) > 0$; biases only shift the gate density and are absorbed into c through Assumption 4.2. □

We isolate what is imported and what is new. The fixed-x lower bound is the recursion of [5, 1] together with the decoupling of [3]; the point here is that it holds under Assumption 4.2 with biases and for any $[0, 1]$-slope activation, and that its uniformity over inputs follows from the net plus

Theorem 5.3. The stability estimate itself is where the submatrix operator-norm bracket enters, and it is new in the two-sided form we need.

5.3. **Stability: few gates flip.** Along a short segment the gate pattern changes only at the neurons whose preactivation is near zero, and there are few of these because Gaussian preactivations anticoncentrate. The Jacobian change is then an operator-norm perturbation supported on a thin gate, bounded by the max-over-thin-patterns statistic of Proposition 3.5 at the small density of flipped neurons.

Theorem 5.3 (Gradient stability)**.** *Under the hypotheses of Proposition 5.2, there is a constant $C = C(L)$ such that w.h.p. over the weights, for all inputs x on the sphere and all x' with $\|x'-x\| \le r$,*

$$\|\nabla F(x') - \nabla F(x)\| \;\le\; C\,\|w\|\left(\tfrac{r\sqrt{\bar n}}{\|x\|}\log\tfrac{\|x\|}{r\sqrt{\bar n}}\right)^{1/2}.$$

In particular, for $r \le \|x\|/\sqrt{\bar n}\cdot\eta$ the right side is $C\|w\|(\eta\log(1/\eta))^{1/2}$, which is $o(\|w\|)$ as $\eta\to 0$; the stability radius needed for the attack, $r \asymp \|x\|/\sqrt{n_L}$ up to logarithms, gives a change small compared with the lower bound $c_g\|w\|$ of Proposition 5.2.

Proof sketch; details in Appendix D. Write the Jacobian difference by the telescoping identity

$$\nabla F(x')^\top - \nabla F(x)^\top = \sum_{\ell=1}^{L} w^\top D_L(x')W_L\cdots\big(D_\ell(x') - D_\ell(x)\big)W_\ell\cdots D_1(x)W_1,$$

so it suffices to bound each $\|(D_\ell(x') - D_\ell(x))W_\ell(\cdots)\|$. The gate difference $D_\ell(x') - D_\ell(x)$ is supported on the neurons whose slope changed between x and x', i.e. those with a preactivation sign change; a neuron flips only if its preactivation at x is within $O(\|z_\ell(x') - z_\ell(x)\|_\infty)$ of zero. Since the preactivations are Gaussian of scale $\varsigma_\ell \asymp \|h_{\ell-1}\|/\sqrt{n_{\ell-1}} \asymp \|x\|/\sqrt{n_{\ell-1}}$ and the layer map is $O(1)$-Lipschitz, the number of flipped neurons at layer ℓ is, by the anticoncentration bound $\mathbb{P}(|N(0,\varsigma^2)| \le \tau) \le \tau/\varsigma$, at most $n_\ell\cdot O(r\sqrt{n_{\ell-1}}/\|x\|) =: \delta_\ell n_\ell$ w.h.p., with $\delta_\ell = O(r\sqrt{\bar n}/\|x\|)$. The factor $(D_\ell(x') - D_\ell(x))W_\ell$ is then a Gaussian block on a pattern of density δ_ℓ, whose operator norm is at most $2\sqrt{\kappa(\delta_\ell)} + o(1)$ by Proposition 3.5, and $\kappa(\delta) \le C\delta\log(1/\delta)$ for small δ (the top-δ mass of a Gaussian). The remaining factors are $O(1)$ in operator norm by Theorem 5.1, and $\|w^\top D_L\cdots\|$ contributes $\|w\|$; multiplying and summing over ℓ gives the stated bound, with the logarithm from $\kappa(\delta_\ell) \le C\delta_\ell\log(1/\delta_\ell)$. Uniformity over x, x' is again by a net plus the $O(1)$-Lipschitz dependence of every factor. □

The three estimates combine into the input of the constructive attack: a small output (next section), a gradient bounded below by $c_g\|w\|$ (Proposition 5.2), and a gradient stable to within $o(\|w\|)$ over the attack segment (Theorem 5.3). The compatibility of these three is what Section 7 verifies.

6. Small outputs

The constructive attack needs the network output to be small relative to the gradient. There are two routes to this, and the paper uses both because they cover different regimes. The first is purely probabilistic and assumes nothing about the symmetry of the network or the input law: a random linear readout makes the output small at every fixed input, and uniformly over inputs in the wide regime. The second is the balance-and-concentration mechanism, which needs the classifier's sign to be balanced but then applies to arbitrary — in particular trained — classifiers on the sphere. We state both; the fragility theorem is proved once and fed by either.

6.1. **Random readout.**

Proposition 6.1 (Small output from a random readout)**.** *Let $F(x) = \langle w, h_L(x)\rangle$ with $w \sim N(0, I_{n_L}/n_L)$ independent of the other weights. For each fixed x and $\delta \in (0,1)$, with probability at least $1-\delta$ over w,*

$$|F(x)| \;\le\; \|h_L(x)\| \sqrt{\tfrac{2}{n_L}\log\tfrac{2}{\delta}}.$$

In the wide regime $d\log\bar{n} = o(n_L)$, w.h.p. over all the weights this holds simultaneously for every x on the sphere of radius R with δ replaced by an exponentially small quantity, so that

$$|F(x)| \;\le\; C\,\|h_L(x)\| \sqrt{\tfrac{d\log\bar{n}}{n_L}} \;\le\; C'\,\|x\| \sqrt{\tfrac{d\log\bar{n}}{n_L}} \qquad \text{for all } x.$$

Proof. Conditionally on $h_L(x)$, $F(x) = \langle w, h_L(x)\rangle \sim N(0, \|h_L(x)\|^2/n_L)$, and a Gaussian tail gives $\mathbb{P}(|F(x)| > \|h_L\|\sqrt{2\log(2/\delta)/n_L}) \le \delta$, which is the fixed-$x$ statement. For the uniform statement, take an ε-net $\mathcal{N}$ of the input sphere of radius R, of size $|\mathcal{N}| \le (3R/\varepsilon)^d$, and choose the threshold $\tau = \|h_L\|_{\max}\sqrt{2(d\log(3R/\varepsilon)+1)/n_L}$ where $\|h_L\|_{\max} = \sup_x \|h_L(x)\| \le C\|x\|$ by the upper half of (8). The union bound gives $\mathbb{P}(\exists\, x_j \in \mathcal{N} : |F(x_j)| > \tau) \le (3R/\varepsilon)^d \cdot 2e^{-\tau^2 n_L/(2\|h_L\|^2_{\max})} \le 2e^{-1}$, and iterating the argument with $d\log(3R/\varepsilon)$ replaced by $d\log(3R/\varepsilon)+s$ makes the failure probability e^{-s}, so w.h.p. $|F(x_j)| \le \tau$ for all net points. Finally F is Lipschitz in x with constant $\|w\|\prod_\ell \|W_\ell\|_{\mathrm{op}} = O(1)$ at fixed depth, so choosing $\varepsilon = R/n$ extends the bound from the net to all x at the cost of $O(\|x\|/n) \ll \tau$, and $\log(3R/\varepsilon) = \log(3n) = O(\log n)$. Hence $|F(x)| \le C\|x\|\sqrt{(d\log n)/n_L}$ for all x w.h.p. This is the union over the net at the net-cardinality scale, not a single exponentially small δ; the $\sqrt{d\log n}$ factor is the price of the union. □

Proposition 6.1 is what removes the oddness and bias-free hypotheses of [1]: no symmetry of the network is used, the biases are arbitrary, and the activation is any $[0,1]$-slope function. The price is that the readout must be random (or, for a trained network, close to a random draw in the relevant inner products); the balance route below removes even that, at the cost of a symmetry hypothesis on the input law.

6.2. **Balance and spherical concentration.**

The second route is the geometric mechanism of [14, 1], which we restate because the fragility theorem uses it for the trained-network regime and because the certification results of Section 8 refer to it. Recall the Lévy constant $C_{\mathrm{L}} = \sqrt{\pi/8}$ and set, for a p-balanced sign (meaning $\min(\mu(F>0), \mu(F<0)) \ge p$),

$$r(p) = \frac{R}{\sqrt{d-1}}\sqrt{2\log(1/p)}, \qquad \varepsilon_{\mathrm{adv}}(p,\delta) = r(p) + \frac{R}{\sqrt{d-1}}\sqrt{2\log(2C_{\mathrm{L}}/\delta)}.$$

Proposition 6.2 (Small output from balance; [1, Thm 2.1])**.** *Let $f : \mathbb{S}^{d-1}(R) \to \mathbb{R}$ be L-Lipschitz with $\operatorname{sgn} f$ p-balanced. Then for every $\delta \in (0,1)$, on a set of μ-measure at least $1-\delta$,*

$$|f(x)| \;\le\; L\,\varepsilon_{\mathrm{adv}}(p,\delta).$$

If f is built from bias-free linear maps, odd activations, and residual additions, then $f(-x) = -f(x)$ and the sign is exactly $\frac{1}{2}$-balanced under any origin-symmetric input law with $\mu(f=0) = 0$.

The two routes are complementary. The random-readout bound is pointwise (every x) and needs no symmetry, but wants a random readout; the balance bound is measure-theoretic (all but a δ fraction) and needs a balanced sign, but then covers any Lipschitz classifier, including one obtained by training, and any origin-symmetric input distribution. In the wide random regime both give $|F(x)| = O(\|x\|/\sqrt{n})$ up to logarithms and a $\sqrt{d}$ factor for the uniform version. We use the random-readout route for the clean every-x statement in Section 7 and note the balance route as the trained-network version in Section 11.

Remark 6.3 (Why small output is the easy half). Neither route uses anything about the gradient; both are consequences of a Lipschitz function being close to its median (concentration) and the median being near zero (balance, or a random readout with mean zero). This is the observation of [1] that the "small output" half of an adversarial argument is generic, and that the architecture-specific work is entirely in the gradient lower bound and its stability — which is exactly what the submatrix theory of Section 5 provides.

7. Universal fragility

We now assemble the master theorem. The engine is the one-step criterion of [1]: if the output is small, the gradient large, and the gradient stable along the attack segment, then a single gradient step crosses the decision boundary. The submatrix theory of Section 5 supplies the gradient facts, and Section 6 supplies the small output; the content of this section is that the three fit together in the wide Gaussian regime for every architecture in the class, uniformly over inputs, with no further hypothesis.

7.1. The constructive engine.

Proposition 7.1 (One-step crossing; [1, Thm 4.1]). *Let $x_0 \in \mathbb{R}^d$, let $a, C_g > 0$, and let $f \in C^1$ on a neighborhood of the ball $\overline{B}(x_0, 2a/C_g)$ with $f(x_0) \neq 0$. Suppose*

(A1) $|f(x_0)| \le a$;
(A2) $\|\nabla f(x_0)\| \ge C_g$;
(A3) $\|\nabla f(x') - \nabla f(x_0)\| \le C_g/3$ *for all* $\|x' - x_0\| \le 2a/C_g$.

Then $x_1 = x_0 - \operatorname{sgn}(f(x_0))\, \dfrac{2a}{\|\nabla f(x_0)\|^2}\, \nabla f(x_0)$ *satisfies* $\operatorname{sgn} f(x_1) \neq \operatorname{sgn} f(x_0)$ *and* $\|x_1 - x_0\| \le 2a/C_g$.

Proof. Write $\sigma_0 = \operatorname{sgn} f(x_0)$ and $\varphi(s) = \sigma_0 f(x_0 - \sigma_0 s\, \nabla f(x_0))$. By the fundamental theorem of calculus and (A3), for $s \le 2a/\|\nabla f(x_0)\|^2$, $\varphi'(s) = -\langle \nabla f(x_0 - \sigma_0 s \nabla f), \nabla f(x_0)\rangle \le -\|\nabla f(x_0)\|^2 + \frac{C_g}{3}\|\nabla f(x_0)\| \le -\frac{2}{3}\|\nabla f(x_0)\|^2$, using (A2). Integrating to $s^* = 2a/\|\nabla f(x_0)\|^2$ gives $\varphi(s^*) \le \varphi(0) - \frac{2}{3}\|\nabla f(x_0)\|^2 s^* = |f(x_0)| - \frac{4}{3}a \le a - \frac{4}{3}a = -\frac{1}{3}a < 0$, so $\sigma_0 f(x_1) < 0$; the step length is $\|x_1 - x_0\| = 2a/\|\nabla f(x_0)\| \le 2a/C_g$. □

The Taylor and dual-norm refinements of [1] sharpen the constant 6 implicit in the compatibility $a \le C_g^2/(6H)$ to 4 or 2 and give an ℓ_∞ signed-gradient (FGSM) certificate; we use the plain version and note the refinements where relevant.

7.2. The master theorem.

Theorem 7.2 (Universal fragility of wide Gaussian networks). *Let F be a network as in Section 4 — fully connected with arbitrary biases, convolutional, a fixed-depth residual network, or a simplified vision transformer (Section 9) — with independent Gaussian weights in the signal-preserving normalization, a random readout $w \sim N(0, I_{n_L}/n_L)$, and any $[0,1]$-slope activation satisfying the nondegeneracy of Proposition 5.2 under Assumption 4.2. Assume the wide regime $dL \log \bar{n} = o(\min_\ell n_\ell)$. Then there is a constant $C = C(L, c_0, c_1, c)$ such that, with high probability over the weights,* simultaneously for every input x on the sphere of radius R *with $F(x) \neq 0$, the single gradient step of Proposition 7.1 applied to F changes the sign of F, with Euclidean length*

$$\|x_1 - x\| \;\le\; C\,\|x\|\,\sqrt{\frac{d \log \bar{n}}{n_L}}.$$

For any fixed input x the length improves to $C\,\|x\|/\sqrt{n_L}$ (no $d \log \bar{n}$ factor).

Proof. Fix the good event, of probability $1 - o(1)$, on which the conclusions of Propositions 5.2 and 6.1 and Theorem 5.3 all hold uniformly over the sphere. On this event, for every x:

(A1) $|F(x)| \le a := C_1 \|x\| \sqrt{(d \log \bar{n})/n_L}$ by Proposition 6.1;

(A2) $\|\nabla F(x)\| \geq C_g := c_g\|w\|$ by Proposition 5.2, and $\|w\| = 1 + o(1)$ w.h.p.;
(A3) for $\|x' - x\| \leq 2a/C_g$, Theorem 5.3 gives $\|\nabla F(x') - \nabla F(x)\| \leq C\|w\|(\eta \log(1/\eta))^{1/2}$ with $\eta \asymp (2a/C_g)\sqrt{\bar{n}}/\|x\| \asymp \sqrt{(d \log \bar{n})\, \bar{n}/n_L}\,/\, c_g$. In the wide regime this is $o(\|w\|) = o(C_g)$, so (A3) holds with room to spare once $\bar{n}\, d \log \bar{n} = o(n_L\, c_g^2)$, which is the wide regime for the readout width; for square-width networks $\bar{n} \asymp n_L$ we use the sharpened stability radius, see Remark 7.3.

Proposition 7.1 then applies at every x, and the step length is $\|x_1 - x\| \leq 2a/C_g = C\|x\|\sqrt{(d \log \bar{n})/n_L}$. For a single fixed x the net is unnecessary and Proposition 6.1 gives $a = C_1\|x\|/\sqrt{n_L}$ directly, so the length is $C\|x\|/\sqrt{n_L}$. □

Remark 7.3 (Width condition). The clean statement asks $\bar{n}\, d \log \bar{n} = o(n_L)$ for the stability step to dominate uniformly. When all widths are comparable, $\bar{n} \asymp n_L \asymp n$, the naive form of (A3) is borderline; the Taylor-remainder version of the engine ([1, Thm 4.6], compatibility constant 2 and only the attack-line curvature) needs only $\|\nabla F(x') - \nabla F(x)\| = o(\|\nabla F\|)$ *along the segment*, which the directional form of Theorem 5.3 gives whenever $d \log n = o(n)$. So the theorem holds in the standard wide regime $d \log n = o(n)$; we state the operator-norm form for transparency.

Remark 7.4 (The scale). For a fixed input the radius $C\|x\|/\sqrt{n_L}$ matches the $\tilde{O}(\|x\|/\sqrt{d})$ of [6] when the readout width tracks the input dimension, and the $c\sqrt{\log(1/\delta)/d}$ of [2], but with three differences: the activation is any $[0, 1]$-slope function rather than ReLU only; the biases are arbitrary; and the constant in the depth dependence is controlled by Proposition 5.2 rather than left exponential. The uniform-over-inputs version, with its extra $\sqrt{d \log \bar{n}}$, is the price of "every x at once" and is what the arrangement bound of Lemma 4.3 makes affordable.

7.3. **Multiclass.** For a multiclass network $F(x) = \arg\max_j \langle w_j, h_L(x)\rangle$ the attack targets the margin against a fixed competitor.

Theorem 7.5 (Multiclass fragility)**.** *Under the hypotheses of Theorem 7.2 with K classes and independent random readouts $w_j \sim N(0, I/n_L)$, w.h.p. simultaneously for every x on the sphere with predicted class $j = F(x)$ there is a competitor $c \neq j$ and a single gradient step on the margin $m_{j,c}(x) = \langle w_j - w_c, h_L(x)\rangle$ of length $\|x_1 - x\| \leq C\|x\|\sqrt{(d \log(Kn))/n_L}$ with $F(x_1) \neq j$.*

Proof. The margin $m_{j,c}$ is a network of the same class with readout $w_j - w_c \sim N(0, 2I/n_L)$, so Proposition 6.1 bounds $|m_{j,c(x)}(x)|$ for the runner-up $c(x)$ (the class with the largest $\langle w_c, h_L\rangle$ below j) — indeed for the runner-up the margin is the smallest positive one, hence $\leq C\|x\|\sqrt{(d \log(Kn))/n_L}$ by a union bound over the K competitors — while the gradient lower bound and stability apply to $m_{j,c}$ verbatim. Proposition 7.1 on $m_{j,c}$ then crosses $m_{j,c} = 0$, at which point c overtakes j. Uniformity is by the same net, with the extra $\log K$ absorbed. □

7.4. **Dual norms.** The ℓ_2 step is not the only one: the same three ingredients give an ℓ_∞ (signed gradient) attack, which is the threat model of most empirical work.

Corollary 7.6 (ℓ_∞ fragility)**.** *Under the hypotheses of Theorem 7.2, w.h.p. simultaneously for every x the signed-gradient step $x_1 = x - \operatorname{sgn}(F(x))\, \frac{2a}{\|\nabla F(x)\|_1} \operatorname{sgn}(\nabla F(x))$ changes the sign of F, with $\|x_1 - x\|_\infty \leq 2a/\|\nabla F(x)\|_1$. Since $\|\nabla F(x)\|_1 \geq \|\nabla F(x)\|_2 \geq c_g\|w\|$ and typically $\|\nabla F\|_1 \asymp \sqrt{d}\, \|\nabla F\|_2$, the ℓ_∞ radius is $O\big(\|x\|\sqrt{\log \bar{n}}/(\sqrt{n_L})\big)$ per coordinate.*

Proof. Apply the dual-norm engine [1, Thm 4.7, Cor 4.8] with $\|\cdot\|_{\mathcal{X}} = \ell_\infty$, dual ℓ_1, curvature bounded along the signed-gradient segment by Theorem 5.3; the compatibility $2a\, H_\infty < \|\nabla F\|_1^2$ holds because $a = o(1)$ while $\|\nabla F\|_1 \geq c_g$ and $H_\infty = o(1)$ in the wide regime. □

8. Certificates and the certification gap

Fragility has a robustness companion. The same operator-norm bracket that upper-bounds the network Lipschitz constant gives, at every input, a radius inside which no perturbation can change the output; this is a genuine certificate, valid for all inputs with no exceptional set. Comparing it with the fragility radius of Theorem 7.2 shows that the $1/\sqrt{n}$ scale is sharp in both directions. We then prove the theorem that explains why one cannot do better by enumerating gate patterns: any certificate required to hold over all patterns of a fixed density inherits the worst-case conditioning, which is exponentially small by the product rate of Theorem 3.2.

8.1. The Lipschitz certificate.

Theorem 8.1 (Uniform robustness certificate)**.** *Under the hypotheses of Theorem 7.2 (fragility not required; only the Gaussian weights and the wide regime), let*

$$L_{\mathrm{net}} = \|w\| \prod_{\ell=1}^{L} \big(2\sqrt{\kappa(c_1)}\big)$$

with c_1 the gate-density upper bound of Assumption 4.2. Then w.h.p. F is L_{net}-Lipschitz on the sphere, and consequently, for every input x, no perturbation of Euclidean norm less than $|F(x)|/L_{\mathrm{net}}$ changes the sign of F. This certified radius holds for all *inputs with no exceptional set.*

Proof. For any x, x' the difference $F(x') - F(x)$ is bounded by the operator norm of the Jacobian integrated along the segment; on each cell of the arrangement (Lemma 4.3) the Jacobian is a fixed product of gated blocks, whose operator norm is at most $\|w\| \prod_\ell \max_{|T_\ell|,|S_\ell| \le c_1 n_\ell} \|P_{T_\ell} W_\ell P_{S_\ell}\|_{\mathrm{op}} \le \|w\| \prod_\ell (2\sqrt{\kappa(c_1)} + o(1))$ by Proposition 3.5, uniformly over cells by the union bound (the count is $(C\bar{n})^{dL}$ and each operator-norm bound fails with probability e^{-cn}). Since the gate densities are at most c_1 everywhere by Assumption 4.2, this bounds the Lipschitz constant on every cell, hence on the sphere. A perturbation smaller than $|F(x)|/L_{\mathrm{net}}$ changes F by less than $|F(x)|$, so cannot change its sign. □

The subgaussian version, with an absolute constant $C(K)$ in place of $2\sqrt{\kappa(c_1)}$, holds for any i.i.d. subgaussian weights of parameter K, by the subgaussian operator-norm bracket of Appendix C; this is what makes the certificate meaningful for non-Gaussian and lightly-trained weights.

Corollary 8.2 (Sharpness of the $1/\sqrt{n}$ scale)**.** *Under the hypotheses of Theorem 7.2 with a random readout, for a fixed input x (equivalently, for all but a vanishing fraction of the sphere, excluding a neighborhood of the decision boundary where $|F(x)|$ is atypically small), w.h.p. the certified radius and the fragility radius bracket each other:*

$$\frac{c\,\|x\|}{\sqrt{n_L}\,L_{\mathrm{net}}} \;\le\; \big(\textit{largest certified radius at } x\big) \;\le\; \big(\textit{attack radius at } x\big) \;\le\; \frac{C\,\|x\|\,\sqrt{d\log\bar{n}}}{\sqrt{n_L}}.$$

Both scale as $\|x\|/\sqrt{n_L}$ up to the constants $L_{\mathrm{net}} = (2\sqrt{\kappa})^L$ and $\prod\sqrt{\check{\kappa}}$; the gap between them is $L_{\mathrm{net}}/\prod_\ell \sqrt{\check{\kappa}(\alpha_\ell)}$, a product of the two-sided gate constants and no power of n.

Proof. The lower bound is Theorem 8.1 with $|F(x)| \ge c\|x\|/\sqrt{n_L}$, which holds w.h.p. at a fixed x because $F(x) \sim N(0, \|h_L\|^2/n_L)$ is not atypically small (anticoncentration of a Gaussian); the upper bound is Theorem 7.2. The ratio of the two radii is $L_{\mathrm{net}} \cdot (\text{attack constant}) = (2\sqrt{\kappa})^L \prod \sqrt{\check{\kappa}}^{-1}$ up to logarithms, independent of n. □

8.2. The certification gap.

The certificate of Theorem 8.1 is loose by the factor $L_{\mathrm{net}}/\prod\sqrt{\check{\kappa}}$ because it uses the worst-case operator norm at every step. One might hope to sharpen it by certifying each gate pattern separately, using the actual conditioning of the realized Jacobian rather than the worst pattern's. The product rate shows this cannot help beyond a point: a certificate valid

over all density-α patterns must survive the worst-conditioned one, and that one is exponentially singular.

Theorem 8.3 (Certification gap)**.** *Consider certifying a depth-k gated linear network $x \mapsto D_k W_k \cdots D_1 W_1 x$ with Gaussian weights, by any method whose guarantee must hold uniformly over all gate tuples $(D_1, \dots, D_k)$ of density α. The smallest conditioning such a method must tolerate is the least singular value of the worst pattern, $M_N^{(k)}(\alpha)$, which by Theorem 3.2 is at most $e^{-2H(\alpha)N/d_{\mathbb{F}}+o(N)}$ w.h.p., and equal to $e^{-(k+1)H(\alpha)N/d_{\mathbb{F}}+o(N)}$ at the sharp (numerically determined) product rate. Consequently any pattern-uniform margin- or inversion-based certificate degrades relative to the typical-pattern conditioning $N^{-(k+1)/2}$ by a factor at least $e^{2H(\alpha)N/d_{\mathbb{F}}+o(N)}$, and by $e^{(k+1)H(\alpha)N/d_{\mathbb{F}}+o(N)}$ at the sharp rate: the union that a uniform certificate performs is dominated by the single worst pattern, whose conditioning sets the rate.*

Proof. A certificate uniform over the $\binom{N}{\alpha N}^{k+1} = e^{(k+1)H(\alpha)N+o(N)}$ patterns must in particular be valid for the pattern minimizing $\sigma_{\min}$, whose value is $M_N^{(k)}(\alpha)$; by Theorem 3.2 this is $\le e^{-2H(\alpha)N/d_{\mathbb{F}}+o(N)}$ w.h.p. (the proved achievability of Proposition B.2), and $= e^{-(k+1)H(\alpha)N/d_{\mathbb{F}}+o(N)}$ at the sharp rate. A margin certificate scales the certified radius by the least singular value of the map it inverts, so its worst-case radius is smaller than the typical-pattern radius (governed by the fixed-support hard edge $N^{-(k+1)/2}$, Proposition 3.6 and its least-sv companion) by the ratio $N^{-(k+1)/2}/M_N^{(k)}$, which is at least $e^{2H(\alpha)N/d_{\mathbb{F}}+o(N)}$ and equals $e^{(k+1)H(\alpha)N/d_{\mathbb{F}}+o(N)}$ at the sharp rate. Either way it is exponential in N. □

Remark 8.4 (Interpretation). Theorem 8.3 is the precise sense in which the adversarial and robust pictures are two sides of one random-matrix law. Fragility is the statement that the *realized* pattern already conditions the network badly enough for a gradient step to work; the certification gap is the statement that no pattern-agnostic certificate can be sharp, because the *worst* pattern is exponentially worse than the realized one, at a rate that is at least $2H(\alpha)/d_{\mathbb{F}}$ unconditionally and $(k+1)H(\alpha)/d_{\mathbb{F}}$ at the sharp product rate of Theorem 3.2. The only way to beat the gap is to certify at the realized pattern — i.e. to use the actual input, which is what a per-input Lipschitz-of-the-realized-Jacobian certificate does; that recovers the $1/\sqrt{n}$ scale of Corollary 8.2 but is not uniform over patterns, and hence not robust to a gate flip, which is exactly what an adversary induces.

8.3. **Certified global robustness of sparse gates.** When the gate density is small enough that even the optimized product norm contracts, the network is robust everywhere, with no small-perturbation attack possible.

Theorem 8.5 (Sparse-gate robustness)**.** *Suppose Assumption 4.2 holds with $c_1 < \alpha^* = 0.042672$ (the root of $\kappa(\alpha^*) = 1/4$). Then w.h.p. the network Lipschitz constant $L_{\text{net}} = (2\sqrt{\kappa(c_1)})^L \|w\|$ tends to 0 as $L \to \infty$, and for $L \ge L_0(c_1)$ the network computes a function whose oscillation on the sphere is $o(1)$: no input has a certified radius smaller than $\Omega(1)$ relative to the output scale, so the $1/\sqrt{n}$ fragility of Theorem 7.2 is absent. Concretely, every perturbation of norm below $|F(x)|/(2\sqrt{\kappa(c_1)})^L$ is harmless, and this radius grows with depth.*

Proof. By Proposition 3.8, $\kappa(c_1) < 1/4$ makes $(2\sqrt{\kappa(c_1)})^L \to 0$, so $L_{\text{net}} \to 0$ and Theorem 8.1 gives a certified radius $|F(x)|/L_{\text{net}}$ that grows geometrically in L. The oscillation bound is the same operator-norm estimate applied to $F(x) - F(x')$ over the whole sphere. □

This is the positive counterpart of the fragility theorems: sparsity of activation, quantified by the gate density falling below α^*, certifies global robustness, and the threshold is exactly the contraction threshold of the optimized submatrix product. It is consistent with the converse of [2], that networks deep enough to be near-constant are not fragile, and sharpens it to a density condition with an explicit constant.

9. Architectures

The master theorem was stated for a generic feedforward network; here we verify the two structural hypotheses it uses — the uniform gain bracket and the gradient lower bound with stability — for convolutional, residual, and transformer networks. In each case the point is that the input–output Jacobian is again a product of gated Gaussian blocks, possibly structured, to which the estimates of Section 5 apply, and that the biases and the choice of $[0,1]$-slope activation change only the gate densities.

9.1. Convolutional networks.

A convolutional layer applies a Gaussian filter bank followed by a gate. With c channels and spatial size s the layer acts on $\mathbb{R}^{cs}$ by a block matrix whose blocks are (multichannel) circulant or Toeplitz Gaussian matrices; the gate keeps the active pixel-channels. The Jacobian is therefore a product of gated structured-Gaussian blocks.

Theorem 9.1 (Convolutional fragility)**.** *Let F be a convolutional network with independent Gaussian filters, arbitrary biases, and any $[0,1]$-slope activation, in the wide regime $d\log(cs) = o(cs)$ (effective width cs). Then the conclusions of Theorem 7.2 and Theorem 8.1 hold: w.h.p. simultaneously for every input a single gradient step of length $O(\|x\|\sqrt{(d\log cs)/(cs)})$ crosses the boundary, and every input has a certified radius of the same order.*

Proof sketch. Two ingredients change relative to the fully connected case. First, the operator-norm bracket of Proposition 3.5 must be applied to structured Gaussian blocks; the upper bound survives because the circulant/Toeplitz structure only decreases the operator norm relative to a full Gaussian of the same variance (the filter has $O(1)$ taps), and the lower bound survives because a fixed input direction is mapped to a Gaussian vector whose top/bottom pixel-channels carry $\kappa, \check{\kappa}$ of its mass as before. Second, the gradient lower bound uses the nondegeneracy recursion of Proposition 5.2 channelwise; this is exactly the content of [5] for C^2 activations, and the $[0,1]$-slope extension is by the convex-hull reduction of Lemma C.3. We do not use any Fourier diagonalization of the nonlinear network — only operator-norm bounds, which the frequency structure of convolutions makes no worse — so the barrier of [1, Rem 10.2] does not obstruct the argument. The stability lemma Theorem 5.3 applies verbatim, with the anticoncentration of the pixel-channel preactivations replacing that of the fully connected preactivations. □

9.2. Residual networks.

A fixed-depth residual network has blocks $h_\ell = h_{\ell-1} + t^{-1/2}\sigma(\Lambda_\ell h_{\ell-1})$ with Gaussian Λ_ℓ and the standard residual scaling. Its block Jacobian is $A_\ell = I + t^{-1/2}D_\ell\Lambda_\ell$, a gated near-identity, and the product $\prod_\ell A_\ell$ has least and largest singular values pinched around 1.

Theorem 9.2 (Residual fragility, ReLU and general slopes)**.** *Let F be a fixed-depth residual network with t blocks, Gaussian Λ_ℓ with $\|\Lambda_\ell\|_{\mathrm{op}} \le M$, any $[0,1]$-slope activation, and arbitrary biases, with $t > S_1^2M^2$ (so the near-identity is invertible), where $S_1 = \|\sigma'\|_\infty \le 1$. Set $\nu = 1 - S_1M/\sqrt{t}$, $\alpha = 1 + S_1M/\sqrt{t}$. Then w.h.p. $\sigma_{\min}(J_h) \ge \nu^t$ and $\|J_h\|_{\mathrm{op}} \le \alpha^t$ uniformly over inputs, and the conclusions of Theorems 7.2 and 8.1 hold with $C_g = \nu^t\|w\|$ and $L_{\mathrm{net}} = \alpha^t\|w\|$.*

Proof sketch. The deterministic block bounds $\sigma_{\min}(A_\ell) \ge \nu$, $\|A_\ell\|_{\mathrm{op}} \le \alpha$ are [1, Prop 5.6]; they hold for any $[0,1]$-slope σ because they use only $\|D_\ell\|_{\mathrm{op}} \le S_1 \le 1$ and the gate support, not the sign structure. What [1] could not do for ReLU was the gradient lower bound, which it obtained only in expectation and only for odd σ via a cross-term cancellation. Here the lower bound $\sigma_{\min}(J_h) \ge \nu^t$ is deterministic given the weights, so it holds for ReLU and every $[0,1]$-slope activation with no oddness: the near-identity structure makes each block invertible with $\sigma_{\min} \ge \nu > 0$ regardless of the gate. The stability bound uses the same telescoping as Theorem 5.3, with the residual scaling $t^{-1/2}$ making the per-block Hessian $H_{\mathrm{res}} = S_2M^2\sqrt{t}\,\alpha^{2t-2}$ ([1, Prop 5.6(c)]), and the compatibility $a \le \nu^{2t}/(6H_{\mathrm{res}}\alpha^t)$ holds in the wide regime because $a = O(\|x\|/\sqrt{n_L}) \to 0$. □

The residual case is the one where the paper most directly strengthens [1]: it removes the oddness and bias-free restrictions and covers ReLU, precisely because the gradient lower bound is now deterministic (near-identity invertibility) rather than an odd-activation expectation.

9.3. **Vision transformers.** A simplified vision transformer, in the sense of [1, §5.3] (bias-free linear maps, no positional embedding, affine-free normalization, odd MLP activation), has a Jacobian that is a product of near-identity residual blocks whose linear part is a gated value-path plus a low-rank attention correction. The attention weights are a softmax of a bilinear form in the input, and the difficulty that [1] left open (Remark 5.20 there) was that at standard temperature the softmax logits are order one, so the attention Jacobian does not concentrate near a fixed linear map. In the wide regime the value-path constants do concentrate, which lets us discharge the compact-set hypotheses of their conditional theorem at random initialization.

Theorem 9.3 (Simplified-ViT fragility at random init, wide regime)**.** *Let F be a simplified vision transformer with L layers, h heads, key dimension d_k, token count n_{tok}, and Gaussian (Xavier) weights, in the wide regime $hd_k \to \infty$ with $d\log(hd_k) = o(hd_k)$. Then w.h.p. the value-path operator norms q_ℓ, k_ℓ, m_ℓ of* [1, Thm 5.16] *are $O(1)$, the standard-temperature Jacobian and Hessian bounds Γ_ℓ, Ξ_ℓ there are finite and $O(1)$ uniformly over the Frobenius ball of radius $R \asymp \sqrt{n_{\mathrm{tok}} d}$, and on the local tube of* [1, Rem 5.17] *generated by the attack neighborhoods the conclusions of Theorems 7.2 and 8.1 hold: a single gradient step of Frobenius length $O(R/\sqrt{hd_k})$ crosses the boundary.*

Proof sketch. The constants $q_\ell = \|W_{Q,\ell}\|_{\mathrm{op}}$, $k_\ell = \|W_{K,\ell}\|_{\mathrm{op}}$, $m_\ell = \|W_{V,\ell}W_{O,\ell}\|_{\mathrm{op}}$ of [1, Thm 5.16] are operator norms of Gaussian matrices, hence $O(1)$ w.h.p. by the Bai–Yin edge; substituting into their $\Gamma_\ell = m_\ell(\sqrt{n_{\mathrm{tok}}} + RL_{P,\ell})$, $\Xi_\ell = m_\ell(RH_{P,\ell} + 2L_{P,\ell})$ gives finite $O(1)$ values on $R \asymp \sqrt{n_{\mathrm{tok}} d}$ because $L_{P,\ell}, H_{P,\ell} = O(R/\sqrt{d_k})$ times the universal softmax constants. The near-identity structure ($\nu_{\mathrm{std}} = 1 - \beta\Gamma > 0$ with $\beta = L^{-1/2}$) then gives the gradient lower bound $\|\nabla F\| \geq \nu_{\mathrm{std}}^L \|w\|$ as in the residual case, deterministically given the weights. What remains is the forward invariance of a set containing the attack trajectories; we replace the global compact-set hypothesis of [1, Thm 5.16] by the local tube of their Remark 5.17, whose existence follows from the stability estimate Theorem 5.3 applied to the value-path (the attention correction is $O(\beta)$ and does not move the trajectory out of an $O(R/\sqrt{hd_k})$ tube). This resolves the wide-regime, simplified, standard-temperature case of their open problem; the fully standard transformer (biases, positional embeddings, affine normalization, GELU) remains open, as those break the balance that supplies the small output. □

Remark 9.4. Theorem 9.3 is a partial resolution of [1, Rem 5.20]: it shows the standard-temperature constants are $O(1)$ w.h.p. at random init and that forward invariance can be localized to a tube, so the conditional theorem becomes an unconditional wide-regime one for the simplified architecture. The random readout route of Proposition 6.1 additionally removes the oddness hypothesis on the MLP that their balance argument needed for the small output, at the cost of the readout being random.

10. Depth: phase transitions and the dynamical limit

The gate density controls a phase transition in depth. Above a density threshold the optimized product norm can grow and fragility persists to any depth; below it, the norm contracts and the network becomes globally robust and eventually near-constant. The threshold is the contraction constant of the optimized submatrix product, and the transition is sharp in the density. We also record the weight-tied limit, where one Gaussian matrix is reused across depth and the long-time behaviour is set by the eigenvalue radius of cyclic submatrix products rather than the operator norm.

10.1. **Density thresholds.**

Theorem 10.1 (Depth phase transition)**.** *Consider a depth-L gated Gaussian network with all gate densities equal to α, in the signal-preserving normalization. Let $\alpha^* = 0.042672$ be the root of $\kappa(\alpha) = 1/4$ and, for fixed support, $\alpha_k^* = (k^k/(k+1)^{k+1})^{1/k}$.*

1. *If $\alpha < \alpha^*$, then w.h.p. the optimized Jacobian operator norm $\prod$-bracket contracts, $\max_{S_\bullet} \|J\|_{\mathrm{op}} \le (2\sqrt{\kappa(\alpha)})^L \to 0$, and by Theorem 8.5 the network is globally robust: every input has certified radius growing with depth, and no single-step attack succeeds for large L.*
2. *If $\alpha^* < \alpha < 1$, the operator-norm bracket does not force contraction: the optimized product norm can exceed one, so fragility of the type in Theorem 7.2 is possible at every depth, and the realized-pattern Jacobian is $\Theta(1)$ so the fragility radius stays $\Theta(\|x\|/\sqrt{n})$.*

Proof. Part (1) is Proposition 3.8 and Theorem 8.5. For part (2), at $\alpha > \alpha^*$ the upper bracket $(2\sqrt{\kappa(\alpha)})^L$ exceeds one and grows, so it does not certify contraction; the realized-pattern Jacobian has the signal-preserving $\Theta(1)$ norm of Proposition 5.2, so the gradient lower bound and hence Theorem 7.2 apply at every fixed depth. □

The gap between $\alpha^* = 0.043$ (optimized) and α_k^* (fixed support, e.g. 0.535 at $k = 4$) is the entropy the adversary buys by choosing patterns: a fixed half-support product of length 4 already contracts, while the optimized-support threshold stays near α^*, because the optimizer can find the rare non-contracting pattern. This is the operator-norm face of the certification gap.

10.2. **The weight-tied dynamical limit.** When one Gaussian matrix W is reused at every layer, $x_{t+1} = \sigma(Wx_t)$, the per-step map is a gated copy of W and the long-time behaviour is a question about cyclic products of its submatrix blocks. The relevant facts, from the companion analysis [10], are that the state norm contracts at the fresh-Gaussian rate, that the activation density is $1/2$, and that no expanding direction survives:

Proposition 10.2 (Weight-tied contraction; [10])**.** *Let W be $N \times N$ with i.i.d. $N(0, 1/N)$ entries and $x_{t+1} = \sigma(Wx_t)$ for ReLU σ. For every fixed horizon T and deterministic input, w.h.p. $\|x_t\|^2 \to 2^{-t}$ for $t \le T$ and the gate density is $1/2$. For every fixed $r > 2^{-1/2}$, w.h.p. there is no ReLU ray of multiplier $\ge r$, so every trajectory whose normalized direction converges decays at every rate exceeding $2^{-1/2}$. The sustained gain of a period-p cycle is the eigenvalue radius of the cyclic submatrix product, $2^{-p/2}$ for asymptotically free blocks by Haagerup–Larsen, below one for all $p \ge 2$.*

The relevance to fragility is that the output at depth t is $|F| \asymp \|x_t\| \asymp 2^{-t/2}\|x_0\|$ (random readout) while the Jacobian also contracts, so the ratio — the fragility radius — is set by the competition between the two, and the eigenvalue-radius contraction of Proposition 10.2 shows the Jacobian never expands along a cycle even though its submatrix operator norm exceeds one. The weight-tied network is thus in the robust phase for the sustained (deep, cyclic) part of its dynamics, and the fragility is a finite-horizon phenomenon governed by the transient submatrix products. This is the dynamical-systems reading of the density threshold: the reused-weight map has effective density $1/2 > \alpha^*$, so it is fragile at finite depth, but its cyclic eigenvalue radius is contractive, so it does not sustain an expanding adversarial direction.

Remark 10.3 (Why the singular value and the eigenvalue disagree)*.* The half-by-half submatrix operator norm is $\sqrt{2} > 1$, and its optimized-support maximum is larger still, up to $2\sqrt{\kappa(1/2)} = 1.927$; yet the cyclic eigenvalue radius is $2^{-p/2} < 1$. The gap is the non-normality of the gated product, and it is why bounds on submatrix *singular* values alone cannot decide the infinite-time dynamics — the ReLU sign constraint, which selects a specific nonnegative-cone-feasible pattern, is what collapses the operator-norm bracket to the eigenvalue radius. The same distinction underlies the certification gap: an adversary works with singular values (worst pattern, operator norm), while the sustained dynamics works with eigenvalues (realized cycle, spectral radius).

11. Trained and semi-random networks

The theorems so far assume Gaussian weights. Trained networks are not Gaussian, but empirically their weight matrices are a Gaussian bulk plus a small number of outlier directions, and the balance route to the small output does not need Gaussianity at all. This section records what survives training: a perturbation-stability statement that degrades the Gaussian bounds gracefully under a bounded deviation of the weights, the balance version of the fragility theorem for trained classifiers, and a discussion of the numerical evidence.

11.1. **Perturbation stability.**

Theorem 11.1 (Stability under weight deviation)**.** *Let $W_\ell = W_\ell^0 + E_\ell$ with W_ℓ^0 Gaussian (satisfying the hypotheses of Theorem 7.2) and E_ℓ arbitrary with $\|E_\ell\|_{\mathrm{op}} \le \varepsilon_\ell$. Then:*

1. *the Lipschitz certificate of Theorem 8.1 holds with L_{net} replaced by $\|w\| \prod_\ell (2\sqrt{\kappa(c_1)} + \varepsilon_\ell)$;*
2. *the gradient lower bound of Proposition 5.2 survives, $\|\nabla F(x)\| \ge c_g\|w\| - \sum_\ell \varepsilon_\ell G_\ell$, where G_ℓ is the product of the operator-norm bounds of the other layers, provided the deviation is small, $\sum_\ell \varepsilon_\ell G_\ell < \frac{1}{2} c_g \|w\|$;*
3. *consequently the fragility conclusion of Theorem 7.2 holds for the perturbed network under this smallness condition, with the constants shifted accordingly.*

Proof. (1) Operator norms are subadditive, $\|D_\ell(W_\ell^0 + E_\ell)P\|_{\mathrm{op}} \le \|D_\ell W_\ell^0 P\|_{\mathrm{op}} + \varepsilon_\ell$, so each factor's bound increases by ε_ℓ and the product bound is $\prod_\ell (2\sqrt{\kappa(c_1)} + \varepsilon_\ell)$. (2) Write the Jacobian of the perturbed network as the Gaussian Jacobian plus a sum of L terms, each replacing one W_ℓ^0 by E_ℓ and bounding the other factors by their operator norms; the correction is at most $\sum_\ell \varepsilon_\ell G_\ell$ in norm, which is subtracted from the Gaussian lower bound $c_g\|w\|$. (3) Under the smallness condition the perturbed gradient is still $\ge \frac{1}{2} c_g \|w\|$, and the small output and stability are perturbed by the same $O(\sum \varepsilon_\ell)$ amounts, so Proposition 7.1 applies with the shifted constants. □

The smallness condition is a low-rank/small-norm deviation from the Gaussian bulk, which is what the Marchenko–Pastur analysis of trained weights [17, 13] identifies: the bulk of the singular spectrum stays at the random edge while a few directions carry the learned signal, so $\|E_\ell\|_{\mathrm{op}}$ is the top learned singular value, typically $O(1)$ and not growing with width, while $c_g\|w\|$ and the certificate scale are $\Theta(1)$. The theorem then says training moves the constants but not the $1/\sqrt{n}$ scale.

11.2. **The balance route for trained classifiers.**

For a trained classifier the readout is not random, so the small output must come from balance. If the trained classifier is accurate and its ground-truth labels are balanced, then its sign is balanced under the input measure (pushed to the sphere), and Proposition 6.2 gives the small output for all but a δ-fraction of inputs. The gradient facts, however, need the weights to be near-Gaussian, which is Theorem 11.1. Combining:

Corollary 11.2 (Trained-network fragility, informal)**.** *Let F be a trained classifier on the sphere whose weights are a Gaussian bulk plus a bounded deviation (Theorem 11.1) and whose sign is p-balanced. Then for all but a δ-fraction of inputs a single gradient step of length $O(\varepsilon_{\mathrm{adv}}(p,\delta) \cdot L_{\mathrm{net}}/c_g)$ crosses the boundary.*

This is the trained-regime statement of Theorem 7.2: the small output is measure-theoretic (from balance) rather than pointwise (from a random readout), and the gradient facts hold up to the weight deviation. It is the theorem that the CIFAR experiments of [1] were designed to probe.

11.3. **Numerical evidence from trained networks.**

The experiments of [1] train small CNNs, ResNets, and ViTs on CIFAR-10 and CIFAR-100 and measure, along training, the one-step attack success, the predicted radius $2m/\|\nabla m\|$ against the empirical attack radius, and the compatibility ratio. Three of their findings are exactly what the present theory predicts and are worth restating

in its language. First, the empirical attack radius is monotone in the predicted radius $2m/\|\nabla m\|$ and stays below it, on both datasets — the conservative direction, since the prediction is an upper bound from the small-output estimate. Second, the one-step vulnerability signatures appear early and persist after the networks become accurate, so fragility is not a symptom of poor accuracy but a structural property of the map, as the balance-plus-gradient mechanism requires. Third, the transformer shows the strongest signatures despite the lowest accuracy, consistent with the near-identity residual structure making its gradient lower bound most robust. The experiments are diagnostics, not certified radii, and the frozen artifacts are catalogued in [1]; we use them as the trained-regime evidence for the scale that Theorem 7.2 and Corollary 8.2 predict, and add the new numerical verification of the product rate in the next section.

12. Numerical verification of the product rate

The one new random-matrix constant in the paper — the product least-singular-value rate $(k+1)H(\alpha)/d_{\mathbb{F}}$ of Theorem 3.2 — differs from the previously conjectured $2H(\alpha)/d_{\mathbb{F}}$ starting at $k=2$, so it is worth checking directly. We do so by exhaustive minimization over all support tuples at small N, where the exponent can be read from a single matrix, and by the cascade construction at slightly larger N, where the achieving pattern can be built but exhaustive search is infeasible.

12.1. **Setup.** For $N \in \{8, 10\}$, density $\alpha = 1/2$ (so $n = N/2$), and real Gaussian entries $N(0, 1/N)$, we compute the single-layer minimum $M_N^{(1)} = \min_{|S_0|=|S_1|=n} \sigma_{\min}(A_1[S_1, S_0])$ and the depth-two minimum $M_N^{(2)} = \min_{S_0,S_1,S_2} \sigma_{\min}(A_2[S_2, S_1]A_1[S_1, S_0])$ over all support triples, reporting the rates $\rho_k = -N^{-1} \log M_N^{(k)}$. For $N = 8$ the triple minimization is over 70^3 tuples and is done exactly; for $N = 10$ it is over 252^3 tuples, so there we report the greedy cascade (the two-sided-optimal first factor followed by the one-sided-optimal second factor), which is the achieving construction of Appendix B and upper-bounds the exhaustive minimum. The discriminating statistic is the ratio ρ_2/ρ_1: Theorem 3.2 predicts $3H/2H = 1.5$, the earlier conjecture $2H/2H = 1$.

12.2. **Results.** The measured rates are as follows (means over 8 trials at $N = 8$, 4 at $N = 10$; $2H(1/2) = 1.386$, $3H(1/2) = 2.079$).

	ρ_1 (exhaustive)	ρ_2 (exhaustive)	ρ_2/ρ_1
$N = 8$	1.375	2.087	1.53
$N = 10$	1.579	2.338 (cascade)	1.50 (cascade)

At $N = 8$, where the depth-two minimum is exact, $\rho_2 = 2.087$ sits on the predicted $3H = 2.079$, and the ratio $\rho_2/\rho_1 = 1.53$ is at the value 1.5 that Theorem 3.2 requires, cleanly separated from the conjectured 1.0. The greedy cascade tracks the exhaustive minimum from above (its rate is a lower estimate of the true rate, and lies within finite-N error), confirming that the joint-support optimization of Appendix B is the mechanism realizing the minimum. At $N = 10$ the cascade ratio is 1.50. The single-layer rate ρ_1 fluctuates around $2H = 1.386$ at these small sizes (across seeds, means of 1.42, 1.31, 1.36 at $N = 8, 10, 12$), the $o(N)$ correction to a first-passage exponent being of either sign; the ratio ρ_2/ρ_1, which cancels the leading correction, is the robust quantity and confirms the linear-in-k growth of the exponent.

12.3. **The anisotropic cascade step.** The first moment of the cascade construction (Proposition B.4) predicts that the least singular value of an anisotropic row ensemble M with covariance Σ of least eigenvalue $\lambda_{\min}$, minimized over row supports, scales as $\sqrt{n\lambda_{\min}}$ times the isotropic rate. We check the $\sqrt{\lambda_{\min}}$ prefactor directly. Taking $\Sigma = \mathrm{diag}(1, \ldots, 1, \varepsilon^2)$ and computing the exact minimum over row supports at $N = 10$, $n = 5$, the minimum is proportional to ε once ε is small enough that the least eigenvalue dominates: the ratio $\min / (\varepsilon \cdot \min|_{\varepsilon=1})$ saturates to a constant

for $\varepsilon \le 0.1$, and the log–log slope of min against ε is 1.0–1.15 there, matching the predicted linear dependence. This is the regime the cascade uses, where $\lambda_{\min} = \sigma_{\min}(Q)^2$ is exponentially small.

The code is `numerics/verify_product_rate.py` and `verify_anisotropic_onesided.py`; both use only `numpy`. The verification refutes the product conjecture of [16] and supports Theorem 3.2 and the first-moment reduction of Proposition B.4.

13. Discussion

The paper makes one claim in two registers. In random-matrix terms: the worst conditioning of a chain of coordinate-optimized Gaussian submatrices decays at the sharp entropy rate $(k+1)H(\alpha)/d_{\mathbb{F}}$, and the optimized operator norm moves only to a bracket set by the retained-mass constants $\kappa, \breve{\kappa}$. In network terms: a wide Gaussian network is fragile at the $1/\sqrt{n}$ scale at every input, robust at exactly the reciprocal scale, and un-certifiable pattern-by-pattern below the same entropy rate — and all three are the same law read through the dictionary that gates are coordinate selections and the Jacobian is a submatrix product.

Three features distinguish the account from its predecessors. It needs no oddness or bias-free hypothesis: the small output comes either from a random readout, which is symmetry-free, or from balance, which covers trained classifiers, and the gradient facts come from submatrix conditioning, which does not see the sign structure. It is two-sided: the same bracket that produces the attack produces a matching certificate, so the scale is pinned rather than bounded on one side. And it is uniform over inputs, because the arrangement bound makes the union over realized gate patterns affordable in the wide regime. The cost is the wide regime itself, $dL \log \bar{n} = o(\min_\ell n_\ell)$, and the Gaussian (or subgaussian, or Gaussian-plus-bounded-deviation) weights; the fully trained, proportional-width regime is reached only through the perturbation-stability theorem, which keeps the scale but not the constants.

Several questions remain. The product rate is proved for equal densities and square bottlenecks; the unequal-density cascade of Remark 3.4 should give the general rate, but the sharp constant for a genuine rectangular bottleneck chain is open, as is the corresponding operator-norm edge for unequal densities [16]. The transformer theorem covers the simplified, standard-temperature architecture at random init in the wide regime; the fully standard transformer, with biases, positional embeddings, affine normalization, and GELU, breaks the balance that supplies the small output, and the random-readout route does not obviously apply because the readout is a learned class token. The certification gap is stated for pattern-uniform certificates; whether a certificate that adapts to the realized pattern but is stable under a single gate flip can beat the $1/\sqrt{n}$ scale is the precise form of the "can we certify robustness" question, and Remark 8.4 suggests the answer is no without controlling the flip, which is what the adversary controls. Finally, the weight-tied dynamical limit shows the eigenvalue radius, not the operator norm, governs the sustained gain; making the infinite-time contraction unconditional requires the multi-time sign-decoupling estimate identified in [10], which is the one genuinely open problem in the reused-weight theory.

The unifying point is methodological. Adversarial fragility has been explained by feature compression [8], by computational hardness [4], and by isoperimetry [12, 1]; the present account locates it in the spectral theory of the network's own Jacobian, and shows that the same spectral object — a product of Gaussian submatrix blocks, optimized over supports — controls the attack, the certificate, and the gap between them, quantitatively and with matching constants. The random-matrix theorems were the missing ingredient; with them the geometry of adversarial examples in deep networks is, in the wide Gaussian regime, a corollary of the conditioning of coordinate-optimized Gaussian maps.

Appendix A. Proof of the two-sided single-layer rate

This appendix proves Theorem 3.1: for an $N\times N$ Gaussian matrix A and $M_N(\alpha)=\min_{|S|=|T|=n}\sigma_{\min}(A[T,S])$, $n=\lfloor \alpha N\rfloor$,

$$-\frac{1}{N}\log M_N(\alpha)\ \xrightarrow{\ \mathbb{P}\ }\ \frac{2H(\alpha)}{d_{\mathbb{F}}}.$$

The upper bound is a first moment; the lower bound is a second moment over an alternately marked family whose correlations are controlled by a sharp two-direction small-ball estimate. The argument is that of [18], reproduced here so the product proof of Appendix B can quote its construction and its Lemma A.2.

A.1. First moment: the upper bound.

Proposition A.1. *For every $\varepsilon>0$, $\mathbb{P}(M_N(\alpha)\le e^{-(2H(\alpha)/d_{\mathbb{F}}+\varepsilon)N})\to 0$.*

Proof. A single $n\times n$ Gaussian matrix G has the hard-edge tail

$$\mathbb{P}(\sigma_{\min}(G)\le t)\ \le\ C_{\mathbb{F}}(\sqrt{n}\,t)^{d_{\mathbb{F}}},\qquad t>0, \tag{9}$$

(real case Edelman and Rudelson–Vershynin; complex analogously, the least squared singular value having bounded density near 0 at scale $1/n$). Each $A[T,S]$ has the law of G, so a union bound over the $\binom{N}{n}^2=e^{2H(\alpha)N+o(N)}$ pairs gives $\mathbb{P}(M_N(\alpha)\le t)\le e^{2H(\alpha)N+o(N)}t^{d_{\mathbb{F}}}$; at $t=e^{-(2H(\alpha)/d_{\mathbb{F}}+\varepsilon)N}$ the right side is $e^{-d_{\mathbb{F}}\varepsilon N+o(N)}\to 0$. □

A.2. The sharp two-direction small-ball estimate.

Let g be a standard $\mathbb{F}$-Gaussian vector in $\mathbb{F}^D$ and u,v unit vectors; write $p(t)=\mathbb{P}(|\zeta|\le t)$ for a standard $\mathbb{F}$-Gaussian scalar. The following is the one analytic ingredient.

Lemma A.2. *For all unit $u,v\in\mathbb{F}^D$ with $|\langle u,v\rangle|^2=R$, $s^2=1-R$, and all $0<t\le 1$,*

$$\mathbb{P}\big(|\langle g,u\rangle|\le t,\ |\langle g,v\rangle|\le t\big)\ \le\ p(t)\,p(2t/s)\ \le\ C_{\mathbb{F}}\,p(t)\,\min\big(1,(t/s)^{d_{\mathbb{F}}}\big). \tag{10}$$

If moreover u,v are independent Haar-random unit vectors in $\mathbb{F}^D$, then averaging over their overlap, for all $D\ge 2$ and $0<t\le 1$,

$$\mathbb{P}\big(|\langle g,u\rangle|\le t,\ |\langle g,v\rangle|\le t\big)\ \le\ C_{\mathbb{F}}\,p(t)^2\,(1+\log(1/t)),$$

with $C_{\mathbb{F}}$ independent of D.

Proof. Put $X=\langle g,u\rangle$, $Y=\langle g,v\rangle$; jointly $\mathbb{F}$-Gaussian, unit variances, $\mathbb{E}[Y\overline{X}]=\rho'$ with $|\rho'|^2=R$. Regress $Y=cX+sW$ with $|c|^2=R$ and W a standard $\mathbb{F}$-Gaussian independent of X. If $|X|\le t$ then $|cX|\le t$, so $|sW|=|Y-cX|\le|Y|+|cX|\le 2t$ on the event; hence the event is contained in $\{|X|\le t\}\cap\{|W|\le 2t/s\}$, and independence of X,W gives $\mathbb{P}\le p(t)p(2t/s)\le C_{\mathbb{F}}p(t)\min(1,(t/s)^{d_{\mathbb{F}}})$ by (4). For the averaged bound, $R=|\langle u,v\rangle|^2\sim\mathrm{Beta}(\frac{d_{\mathbb{F}}}{2},\frac{(D-1)d_{\mathbb{F}}}{2})$, so $s^2=1-R\sim\mathrm{Beta}(\frac{(D-1)d_{\mathbb{F}}}{2},\frac{d_{\mathbb{F}}}{2})$, and with $\alpha_0=d_{\mathbb{F}}/2$, $b_0=(D-1)d_{\mathbb{F}}/2$,

$$\mathbb{E}[\min(1,(t/s)^{d_{\mathbb{F}}})]=t^{d_{\mathbb{F}}}\mathbb{E}[s^{-d_{\mathbb{F}}}\mathbb{1}_{\{s\ge t\}}]+\mathbb{P}(s<t).$$

The first expectation is $\frac{1}{B(b_0,\alpha_0)}\int_{t^2}^1 w^{b_0-\alpha_0-1}(1-w)^{\alpha_0-1}\mathrm{d}w$; at $D=2$ the exponent $b_0-\alpha_0-1=-1$ makes it $\le C_{\mathbb{F}}(1+\log(1/t))$, and at $D\ge 3$ the exponent exceeds -1, the integral converges, and the ratio of Beta functions is bounded uniformly in D ($\Gamma(x+a)/\Gamma(x)\asymp x^a$). The second is $\mathbb{P}(\mathrm{Beta}(b_0,\alpha_0)<t^2)\le C_{\mathbb{F}}t^{2b_0}\le C_{\mathbb{F}}t^{d_{\mathbb{F}}}$ since $2b_0=(D-1)d_{\mathbb{F}}\ge d_{\mathbb{F}}$. Hence $\mathbb{E}[\min(1,(t/s)^{d_{\mathbb{F}}})]\le C_{\mathbb{F}}t^{d_{\mathbb{F}}}(1+\log(1/t))$, and $t^{d_{\mathbb{F}}}\le p(t)/c_{\mathbb{F}}$ finishes. □

The averaging is packaged for reuse as follows; the factor Λ carries polynomial overlap prefactors.

Lemma A.3. *Let $j\ge 1$, $q=d_{\mathbb{F}}j$, and $\sigma^2\in[0,1]$ a random variable with, for some $\beta\ge d_{\mathbb{F}}(j+1)/2$, $\gamma\ge 1$, $\Lambda\ge 1$, $\mathbb{P}(\sigma^2\le x)\le\Lambda(\gamma x)^{\beta}$ for all $x\in(0,1]$. Then for $\sqrt{\gamma}\,t\le 1$, $\mathbb{E}[\min(1,(2t/\sigma)^q)]\le C_{\mathbb{F},j}\Lambda\gamma^{q/2}t^q$.*

Proof. Since $\gamma x \le 1$ where the bound is nontrivial, take $\beta = d_{\mathbb{F}}(j+1)/2$. Then $\mathbb{E}[\min(1,(2t/\sigma)^q)] \le \mathbb{P}(\sigma^2 \le 4t^2) + (2t)^q \mathbb{E}[\sigma^{-q}\mathbb{1}_{\{\sigma^2>4t^2\}}]$; the first term is $\le \Lambda(4\gamma t^2)^\beta \le C\Lambda\gamma^{q/2}t^q$. Integrating the second by parts and splitting at $w_* = (\gamma\Lambda^{1/\beta})^{-1}$, using $\beta - q/2 = d_{\mathbb{F}}/2$, both pieces are $\le C\Lambda\gamma^{q/2}$; multiplying by $(2t)^q$ finishes. □

A.3. **The marked-row construction.** Fix $\varepsilon \in (0, H(\alpha)/d_{\mathbb{F}})$, set $t = t_N = e^{-(2H(\alpha)/d_{\mathbb{F}} - \varepsilon)N}$, and a marked set $M \subseteq [N]$ with $|M| = m_N = \lceil\sqrt{N}\rceil$. A *configuration* is $c = (S, R, y)$ with $|S| = n$; $y \in M$; $R \subseteq [N]\setminus M$, $|R| = n-1$. The rows $\{A_i|_S : i \in R\}$ span a hyperplane in $\mathbb{F}^S \cong \mathbb{F}^n$ a.s.; let $u_c \in \mathbb{F}^S$ be a unit normal. Put $E_c = \{|\langle A_y|_S, u_c\rangle| \le t\}$ and $W = \sum_c \mathbb{1}_{E_c}$.

Lemma A.4. *If $W > 0$ then $M_N(\alpha) \le t$.*

Proof. On E_c, set $T = R \cup \{y\}$, $|T| = n$. For $i \in R$, $\langle A_i|_S, u_c\rangle = 0$; hence $\|A[T,S]u_c\|^2 = |\langle A_y|_S, u_c\rangle|^2 \le t^2$, so $\sigma_{\min}(A[T,S]) \le t$. □

Lemma A.5. $\mathbb{E}W = m_N\binom{N}{n}\binom{N-m_N}{n-1}p(t) = e^{2H(\alpha)N+o(N)}t^{d_{\mathbb{F}}} = e^{d_{\mathbb{F}}\varepsilon N+o(N)} \to \infty$.

Proof. Since $y \notin R$, A_y is independent of u_c, so $\mathbb{P}(E_c) = p(t)$; there are $m_N\binom{N}{n}\binom{N-m_N}{n-1}$ configurations, and $m_N = o(N)$ makes both binomials $e^{H(\alpha)N+o(N)}$. □

A.4. **Second moment and the lower bound.**

Proposition A.6. *For every $\varepsilon > 0$, $\mathbb{P}(M_N(\alpha) > e^{-(2H(\alpha)/d_{\mathbb{F}}-\varepsilon)N}) \to 0$.*

By Lemma A.4 and Chebyshev, $\mathbb{P}(M_N > t) \le \mathbb{P}(W = 0) \le \mathrm{Var}(W)/(\mathbb{E}W)^2$, so it suffices that $\mathrm{Var}(W) = o((\mathbb{E}W)^2)$, i.e. $\mathbb{E}[W^2] \le (\mathbb{E}W)^2(1+o(1))$. Write $\mathcal{G} = \sigma(A_i : i \ne y)$ for a pair with marked rows y, y'.

Distinct marked rows. If $y \ne y'$, then A_y enters neither u_c (built from $R \not\ni y$), nor $u_{c'}$ (from $R' \not\ni y$ since $y \in M$, $R' \subseteq [N]\setminus M$), nor $A_{y'}$; so $E_{c'}$ and u_c are $\mathcal{G}$-measurable while $\mathbb{P}(E_c \mid \mathcal{G}) = p(t)$, giving $\mathbb{P}(E_c \cap E_{c'}) = p(t)^2$. Summed, these pairs contribute at most $(\#\text{configs})^2 p(t)^2 = (\mathbb{E}W)^2$.

Same marked row. If $y = y'$, condition on $\mathcal{G}$; then $u_c, u_{c'}$ are determined and, with $\tilde{u}_c, \tilde{u}_{c'}$ the zero-padded normals, (10) gives $\mathbb{P}(E_c \cap E_{c'} \mid \mathcal{G}) \le C_{\mathbb{F}} p(t)\min(1,(t/s)^{d_{\mathbb{F}}})$, $s^2 = 1-R$, $R = |\langle \tilde{u}_c, \tilde{u}_{c'}\rangle|^2$. Set $D_{\mathrm{c}} = n - |S \cap S'|$, $D_{\mathrm{r}} = n - |R \cap R'|$ (so $D_{\mathrm{r}} \ge 1$). The overlap has two tail bounds.

Lemma A.7. *For all $x \in (0,1]$:* (i) *if $D_{\mathrm{r}} \ge 2$, $\mathbb{P}(R > 1-x) \le C_{\mathbb{F}} x^{(D_{\mathrm{r}}-1)d_{\mathbb{F}}/2}$, uniformly in D_{r}, N;* (ii) *if $D_{\mathrm{c}} \ge 1$, $\mathbb{P}(R > 1-x) \le n(C_{\mathbb{F}} nx/D_{\mathrm{c}})^{D_{\mathrm{c}}d_{\mathbb{F}}/2}$.*

Proof. (ii) $R \le \|u_c|_{S\cap S'}\|^2 = 1 - Z$, $Z = \|u_c|_{S\setminus S'}\|^2$; u_c is Haar on the sphere of $\mathbb{F}^S$, so $Z \sim \mathrm{Beta}(a,b)$, $a = D_{\mathrm{c}}d_{\mathbb{F}}/2$, $b = (n - D_{\mathrm{c}})d_{\mathbb{F}}/2$. Then $\mathbb{P}(Z \le x) \le x^a/(aB(a,b)) \le n(C_{\mathbb{F}}nx/D_{\mathrm{c}})^a$ by $\Gamma(a+b) \le \Gamma(b)(a+b)^{a+1}$ and $\Gamma(a+1) \ge (a/e)^a$. (i) Condition on the rows R (determining u_c); the $D_{\mathrm{r}} - 1$ rows in $R' \setminus R$ project to independent Gaussians in $W_0 = (\mathrm{span}\{A_i|_{S'} : i \in R \cap R'\})^\perp$, of dimension D_{r}, and $u_{c'}$ is Haar on the sphere of W_0 orthogonal to them, i.e. Haar on a D_{r}-sphere; so $R \le |\langle \hat{w}, u_{c'}\rangle|^2 = 1 - Z'$ with $Z' \sim \mathrm{Beta}(b_0, a_0)$, $b_0 = (D_{\mathrm{r}}-1)d_{\mathbb{F}}/2$, $a_0 = d_{\mathbb{F}}/2$, giving $\mathbb{P}(Z' \le x) \le C_{\mathbb{F}} x^{b_0}$. □

Now average $C_{\mathbb{F}}p(t)\min(1,(t/s)^{d_{\mathbb{F}}})$ over three regions, using Lemma A.3 ($j = 1$, $q = d_{\mathbb{F}}$) and the counting facts that the fraction of ordered pairs (S, S') with $D_{\mathrm{c}} \le a$ is $\le N^{2a}e^{-H(\alpha)N+o(N)}$, likewise for rows.

$D_{\mathrm{r}} \ge 3$: the row bound gives $\beta = (D_{\mathrm{r}}-1)d_{\mathbb{F}}/2 \ge d_{\mathbb{F}}$, $\gamma = C_{\mathbb{F}}$, $\Lambda = 1$, so $\mathbb{P}(E_c \cap E_{c'}) \le C_{\mathbb{F}}p(t)t^{d_{\mathbb{F}}} \le C_{\mathbb{F}}p(t)^2$; summed over m_N marked rows, $\le C_{\mathbb{F}}(\mathbb{E}W)^2/m_N = o((\mathbb{E}W)^2)$.

$D_{\mathrm{r}} \le 2$, $D_{\mathrm{c}} \ge 2$: the column bound gives $\mathbb{P}(E_c \cap E_{c'}) \le N^C p(t)^2$; the rows are pinned ($\le N^4 e^{-H(\alpha)N}$ fraction), so the contribution is $N^C e^{-H(\alpha)N+o(N)}(\mathbb{E}W)^2 = o((\mathbb{E}W)^2)$.

$D_{\mathrm{r}} \le 2$, $D_{\mathrm{c}} \le 1$: the crude bound $\mathbb{P}(E_c \cap E_{c'} \mid \mathcal{G}) \le p(t)$ gives ratio $C_{\mathbb{F}}t^{-d_{\mathbb{F}}} = C_{\mathbb{F}}e^{(2H(\alpha)-d_{\mathbb{F}}\varepsilon)N}$ to $p(t)^2$; both coordinates pinned ($\le N^C e^{-2H(\alpha)N}$ fraction), so the contribution is $N^C e^{-d_{\mathbb{F}}\varepsilon N+o(N)}(\mathbb{E}W)^2 = o((\mathbb{E}W)^2)$, including the diagonal.

Hence $\mathbb{E}[W^2] \le (\mathbb{E}W)^2(1+o(1))$, so $\mathbb{P}(W=0) \to 0$ and Proposition A.6 follows. Theorem 3.1 is Propositions A.1 and A.6. □

Remark A.8. The marked set forbids a configuration's normal-defining rows from the marked set, which removes the $\Theta(1)$ fraction of genuinely correlated pairs in which each configuration's marked row is the other's normal row, leaving only the same-marked-row pairs of weight $\Theta(1/m_N)$. Any $m_N \to \infty$, $m_N = o(N)$ works.

Appendix B. The product least singular value

This appendix proves the bracket of Theorem 3.2 and explains the sharp constant. Recall $M_N^{(k)}(\alpha) = \min_{S_0,\dots,S_k} \sigma_{\min}(P_{S_k} A_k \cdots P_{S_1} A_1 P_{S_0})$ with all supports of size $n = \lfloor \alpha N \rfloor$, and set $\rho_k = -N^{-1} \log M_N^{(k)}(\alpha)$. Theorem 3.2 asserts $2H(\alpha)/d_{\mathbb{F}} \le \liminf \rho_k \le \limsup \rho_k \le (k+1)H(\alpha)/d_{\mathbb{F}}$, with the upper end the sharp value. The two ends are Propositions B.1 and B.2; the sharp value and the mechanism are Remark B.3.

B.1. The union bound: $\rho_k \le (k+1)H/d_{\mathbb{F}}$.

Proposition B.1. *For every $\varepsilon > 0$, $\mathbb{P}\big(M_N^{(k)}(\alpha) \le e^{-((k+1)H(\alpha)/d_{\mathbb{F}}+\varepsilon)N}\big) \to 0$; hence $\limsup_N \rho_k \le (k+1)H(\alpha)/d_{\mathbb{F}}$.*

Proof. Fix a support tuple $(S_0,\dots,S_k)$. The compressed product $M = F_k \cdots F_1$, with $F_j = A_j[S_j, S_{j-1}]$ independent $n \times n$ Gaussian blocks, satisfies $\sigma_{\min}(M) \ge \prod_{j=1}^k \sigma_{\min}(F_j)$. Writing $X_j = -\log \sigma_{\min}(F_j) \ge 0$, the hard-edge tail (9) gives $\mathbb{P}(X_j \ge x) = \mathbb{P}(\sigma_{\min}(F_j) \le e^{-x}) \le C_{\mathbb{F}}(\sqrt{n}\, e^{-x})^{d_{\mathbb{F}}} = C_{\mathbb{F}} n^{d_{\mathbb{F}}/2} e^{-d_{\mathbb{F}} x}$, so each X_j has an exponential upper tail of rate $d_{\mathbb{F}}$ and prefactor $a = C_{\mathbb{F}} n^{d_{\mathbb{F}}/2}$. The X_j are independent, so the sum $\Sigma = \sum_{j=1}^k X_j$ has, by the standard convolution bound for exponential tails,

$$\mathbb{P}(\Sigma \ge L) \;\le\; C_k\, a^k\, \frac{L^{k-1}}{(k-1)!}\, e^{-d_{\mathbb{F}} L} \qquad (L \ge k \log a),$$

by stochastic domination of each X_j by a shifted exponential $\frac{\log a}{d_{\mathbb{F}}} + \mathrm{Exp}(d_{\mathbb{F}})$ and the Gamma tail of their sum; this is the only range used, since $L = \Theta(N)$ below. With $L = -\log t$ and $t = e^{-((k+1)H/d_{\mathbb{F}}+\varepsilon)N}$, this is $e^{o(N)} t^{d_{\mathbb{F}}}$ (since $a^k L^{k-1} = e^{o(N)}$ for fixed k, $n \le N$, $L = O(N)$), so $\mathbb{P}(\sigma_{\min}(M) \le t) \le e^{o(N)} t^{d_{\mathbb{F}}}$. A union bound over the $\binom{N}{n}^{k+1} = e^{(k+1)H(\alpha)N + o(N)}$ tuples gives $\mathbb{P}(M_N^{(k)} \le t) \le e^{(k+1)H(\alpha)N + o(N)} t^{d_{\mathbb{F}}} = e^{-d_{\mathbb{F}}\varepsilon N + o(N)} \to 0$. □

B.2. A two-sided-singular factor: $\rho_k \ge 2H/d_{\mathbb{F}}$.

Proposition B.2. *For every $\varepsilon > 0$, $\mathbb{P}\big(M_N^{(k)}(\alpha) > e^{-(2H(\alpha)/d_{\mathbb{F}}-\varepsilon)N}\big) \to 0$; hence $\liminf_N \rho_k \ge 2H(\alpha)/d_{\mathbb{F}}$.*

Proof. By Theorem 3.1 applied to A_1, w.h.p. there are supports S_0, S_1 and a unit vector $u \in \mathbb{F}^{S_0}$ (the least right singular vector of $F_1 = A_1[S_1, S_0]$) with $\|F_1 u\| = \sigma_{\min}(F_1) \le e^{-(2H(\alpha)/d_{\mathbb{F}}-\varepsilon/2)N}$. Extend to any supports $S_2,\dots,S_k$. Then, with $F_j = A_j[S_j, S_{j-1}]$,

$$M_N^{(k)}(\alpha) \;\le\; \sigma_{\min}(F_k \cdots F_1) \;\le\; \|F_k \cdots F_1 u\| \;\le\; \Big(\prod_{j=2}^k \|F_j\|_{\mathrm{op}}\Big) \|F_1 u\|.$$

Each $\|F_j\|_{\mathrm{op}} \le 2\sqrt{n}(1+o(1))$ w.h.p. by Bai–Yin (choosing, say, $S_j = [n]$; here the entries are standard, so the edge is at scale $\sqrt{n}$), so the product of the $k-1$ operator norms is at most $(2\sqrt{n})^{k-1}(1+o(1)) = e^{o(N)}$ for fixed k, and $M_N^{(k)} \le e^{-(2H(\alpha)/d_{\mathbb{F}}-\varepsilon/2)N} e^{o(N)} \le e^{-(2H(\alpha)/d_{\mathbb{F}}-\varepsilon)N}$ w.h.p. The rate is insensitive to the entry normalization, since scaling A_j by a constant shifts $-N^{-1}\log\sigma_{\min}$ by $o(1)$. □

This already improves the lower end of the bracket recorded in [16] from $H(\alpha)/d_{\mathbb{F}}$ (one one-sided-singular factor) to $2H(\alpha)/d_{\mathbb{F}}$ (one two-sided-singular factor): the two-sided construction of Appendix A is strictly better than making a single factor one-sided singular.

B.3. **The sharp constant and why the cascade is not enough.**

Remark B.3 (Achievability of $(k+1)H/d_{\mathbb{F}}$). The numerics of Section 12 determine $\rho_k = (k+1)H(\alpha)/d_{\mathbb{F}}$: at $k=2$, $N=8$ the exhaustive minimum over all support triples has rate 2.087, on the predicted $3H(1/2) = 2.079$, and the ratio $\rho_2/\rho_1 = 1.53$ matches the value 1.5 that $(k+1)H/d_{\mathbb{F}}$ requires and excludes the $2H/d_{\mathbb{F}}$ guess of [16] (ratio 1). The following records what is and is not proved about the achievability of this value.

The upper end $(k+1)H/d_{\mathbb{F}}$ is a first-moment threshold: there are $e^{(k+1)H(\alpha)N}$ support tuples, and a single fixed product $F_k \cdots F_1$, being a product of k independent square Gaussian blocks, has the product hard edge $\mathbb{P}(\sigma_{\min} \le t) = t^{d_{\mathbb{F}}+o(1)}$ (the extra factors contribute logarithms, not a power; Akemann–Ipsen–Kieburg, Forrester), so the expected number of tuples with $\sigma_{\min} \le e^{-((k+1)H/d_{\mathbb{F}}-\varepsilon)N}$ is $e^{(k+1)H(\alpha)N} e^{-((k+1)H(\alpha)-d_{\mathbb{F}}\varepsilon)N} = e^{d_{\mathbb{F}}\varepsilon N} \to \infty$. The mean diverges below the threshold, so a matching second moment gives achievability.

The construction that realizes it is a cascade, but not the naive one. Suppose $S_0, \dots, S_{k-1}$ are already chosen so that the partial product $Q = F_{k-1} \cdots F_1$ (an $n \times n$ matrix on rows S_{k-1}, columns S_0) has $\sigma_{\min}(Q) \le e^{-(kH(\alpha)/d_{\mathbb{F}}-\varepsilon)N}$, with right singular vector $v \in \mathbb{F}^{S_0}$ (so $\|Qv\| = \sigma_{\min}(Q)$). Adjoin the last factor by choosing the rows S_k: the composite $M(S_k) = A_k[S_k, S_{k-1}]\, Q$ has rows $m_i = a_i^{\top} Q$ with $a_i = A_k[i, S_{k-1}]$ i.i.d. standard Gaussian, so $M(\,\cdot\,)$ is the row-selection of an $N \times n$ matrix whose i.i.d. rows have covariance $\Sigma = Q^{\top}Q$ (acting on $\mathbb{F}^{S_0}$). Minimizing $\sigma_{\min}(M(S_k))$ over row supports is therefore the one-sided extremal least singular value problem of [15] for an *anisotropic* Gaussian row ensemble, and the marked-row construction there makes $\sigma_{\min}$ small at the one-sided rate $H(\alpha)/d_{\mathbb{F}}$ times the scale $\sqrt{\lambda_{\min}(\Sigma)} = \sigma_{\min}(Q)$ of the covariance's smallest eigenvalue: Σ inherits the tiny eigenvalue $\sigma_{\min}(Q)^2$ along v, and the row selection adds its own $H(\alpha)/d_{\mathbb{F}}$ of entropy on top. This yields $\sigma_{\min}(M(S_k)) \le e^{-((k+1)H(\alpha)/d_{\mathbb{F}}-\varepsilon')N}$, one more $H(\alpha)/d_{\mathbb{F}}$ per layer, telescoping from the two-sided base $\rho_1 = 2H(\alpha)/d_{\mathbb{F}}$ to $\rho_k = (k+1)H(\alpha)/d_{\mathbb{F}}$. The naive cascade that fixes the carried direction η and steers it — minimizing $\|P_{S_k} A_k \eta\|$ over S_k — gains only the $O(1)$ factor $\sqrt{\check{\kappa}(\alpha)}$ and reaches merely $2H(\alpha)/d_{\mathbb{F}}$; the gain comes instead from re-optimizing the null direction of the whole composite, which the anisotropic one-sided law does.

The first-moment side of this cascade step is rigorous, with no condition on Σ; only the second moment remains. This is the content of Proposition B.4 below, which shows that the anisotropic marked count diverges in mean below the threshold $(k+1)H(\alpha)/d_{\mathbb{F}}$. What obstructs a full proof is the covariance $\Sigma = Q^{\top}Q$ being *exponentially* ill-conditioned — its condition number is $\sigma_{\max}(Q)^2/\sigma_{\min}(Q)^2 = e^{\Theta(N)}$ — so the second-moment concentration cannot be quoted from the subexponential-covariance regime of [15], and must be re-run with the configuration-dependent prefactor identified in Proposition B.4. The numerics of Section 12 confirm the value (the cascade reaches $(k+1)H/d_{\mathbb{F}}$; at $k=2$, $N=8$ the greedy cascade rate is 1.95, approaching $3H = 2.08$ from below while the exhaustive minimum sits on $3H$), and the anisotropic scaling itself is checked directly in `verify_anisotropic_onesided.py`. The bracket of Theorem 3.2 is what is proved; the sharp value is the numerically determined answer inside it.

Proposition B.4 (First moment of the anisotropic cascade step)**.** *Let M be $N \times n$ with i.i.d. rows $m_i \sim N(0, \Sigma)$, $\Sigma \succ 0$ with least eigenvalue $\lambda_{\min}$, and let $\mathcal{M} \subseteq [N]$ be a marked set with $|\mathcal{M}| = \lceil \sqrt{N} \rceil$. For a configuration $c = (R, y)$ with $R \subseteq [N] \setminus \mathcal{M}$, $|R| = n-1$, and $y \in \mathcal{M}$, let u_R be a unit normal to $\{m_i : i \in R\}$ and $E_c = \{|\langle m_y, u_R\rangle| \le t\}$; set $W = \sum_c \mathbb{1}_{E_c}$. Then:*

(i) *$W > 0$ implies $\min_{|T|=n} \sigma_{\min}(M[T, \cdot]) \le t$;*

(ii) *writing ν for the (Haar-distributed) unit normal to $n-1$ i.i.d. standard Gaussian vectors, $\|\Sigma^{1/2}u_R\| = 1/\|\Sigma^{-1/2}\nu\|$ in law, and $\mathbb{P}(E_c \mid \{m_i\}_{i\in R}) = p\big(t\,\|\Sigma^{-1/2}\nu\|\big)$ exactly, so*

$$\mathbb{E}W = \binom{N}{n-1}|\mathcal{M}|\;\mathbb{E}_\nu\big[p\big(t\,\|\Sigma^{-1/2}\nu\|\big)\big];$$

(iii) *consequently $\mathbb{E}W \;\geq\; c\binom{N}{n-1}|\mathcal{M}|\,\dfrac{t}{\sqrt{n\,\lambda_{\min}}}$ for a constant $c = c(\mathbb{F}) > 0$ and all $t \leq \sqrt{n\lambda_{\min}}$, so $\mathbb{E}W \to \infty$ whenever $t \geq \sqrt{n\lambda_{\min}}\,e^{-(H(\alpha)/d_{\mathbb{F}}-\varepsilon)N}$.*

Applied along the cascade with $\Sigma = Q^\top Q$ and $\lambda_{\min} = \sigma_{\min}(Q)^2 = e^{-(2kH(\alpha)/d_{\mathbb{F}}-o(1))N}$, part (iii) gives $\mathbb{E}W \to \infty$ for every $t \geq e^{-((k+1)H(\alpha)/d_{\mathbb{F}}-\varepsilon)N}$: the first moment diverges below the sharp threshold.

Proof. (i) On E_c the row set $T = R \cup \{y\}$ has $|T| = n$, and $\|M[T,\cdot]\,u_R\|^2 = \sum_{i\in T}|\langle m_i, u_R\rangle|^2 = |\langle m_y, u_R\rangle|^2 \leq t^2$ since $\langle m_i, u_R\rangle = 0$ for $i \in R$; as u_R is a unit vector, $\sigma_{\min}(M[T,\cdot]) \leq t$.

(ii) Write $m_i = \Sigma^{1/2}g_i$ with g_i i.i.d. standard Gaussian. The condition $m_i \perp u_R$ for $i \in R$ reads $\langle g_i, \Sigma^{1/2}u_R\rangle = 0$, so $\Sigma^{1/2}u_R$ is parallel to the unit normal ν of $\{g_i : i \in R\}$, which is Haar because the g_i are isotropic. Thus $\Sigma^{1/2}u_R = c\nu$ with $c = \|\Sigma^{1/2}u_R\|$, and $u_R = c\,\Sigma^{-1/2}\nu$; unit norm forces $c = 1/\|\Sigma^{-1/2}\nu\|$. Since $y \notin R$, g_y is independent of u_R, and $\langle m_y, u_R\rangle = \langle g_y, \Sigma^{1/2}u_R\rangle = c\langle g_y, \nu\rangle$ is $N(0, c^2)$ given the defining rows, so $\mathbb{P}(E_c \mid \{m_i\}_{i\in R}) = p(t/c) = p(t\|\Sigma^{-1/2}\nu\|)$. Taking expectations and multiplying by the number $\binom{N}{n-1}|\mathcal{M}|$ of configurations gives the stated mean.

(iii) Diagonalize $\Sigma = \sum_j \lambda_j w_j w_j^\top$ with $\lambda_{\min} = \lambda_n$ the least eigenvalue and eigenvector w_n. Then $\|\Sigma^{-1/2}\nu\|^2 = \sum_j \lambda_j^{-1}\langle\nu, w_j\rangle^2 \geq \lambda_n^{-1}\langle\nu, w_n\rangle^2$, so $\|\Sigma^{-1/2}\nu\| \geq \lambda_n^{-1/2}|\langle\nu, w_n\rangle|$. Using $p(s) \geq c_{\mathbb{F}}\min(s, 1)$ (valid for every standard $\mathbb{F}$-Gaussian small-ball probability) and $t \leq \sqrt{n\lambda_n}$,

$$\mathbb{E}_\nu\big[p(t\|\Sigma^{-1/2}\nu\|)\big] \;\geq\; c_{\mathbb{F}}\,\mathbb{E}_\nu\big[\min\big(t\lambda_n^{-1/2}|\langle\nu, w_n\rangle|,\,1\big)\big] \;\geq\; c'_{\mathbb{F}}\,t\,\lambda_n^{-1/2}\,\mathbb{E}_\nu|\langle\nu, w_n\rangle|,$$

where the last step keeps only the part of the range where the minimum equals its first argument, which has $\Theta(1)$ probability. For Haar ν on $\mathbb{S}^{n-1}$, $\mathbb{E}_\nu|\langle\nu, w_n\rangle| = \Theta(1/\sqrt{n})$, so $\mathbb{E}_\nu[p(t\|\Sigma^{-1/2}\nu\|)] \geq c''\,t/\sqrt{n\lambda_n}$, and (iii) follows from (ii) with $\binom{N}{n-1}|\mathcal{M}| = e^{H(\alpha)N+o(N)}$. □

Remark B.5 (The remaining second moment). By Proposition B.4(i) and the second-moment method, achievability of the cascade step reduces to $\mathrm{Var}(W) = o((\mathbb{E}W)^2)$. Two configurations c, c' with distinct marked rows are, unlike the isotropic case of Appendix A, no longer exactly independent: the conditional probability $p(t\|\Sigma^{-1/2}\nu\|)$ depends on the configuration through its normal ν, so $\mathbb{P}(E_c \cap E_{c'}) = \mathbb{E}[\mathbb{1}_{E_{c'}}p(t\|\Sigma^{-1/2}\nu_c\|)]$ carries a covariance between the prefactor $\|\Sigma^{-1/2}\nu_c\|$ and the event $E_{c'}$ whenever the defining row sets R, R' overlap. This covariance is the sole obstruction: it vanishes for disjoint R, R' and is otherwise controlled by the overlap of the two Haar normals, exactly the quantity the two-direction estimate Lemma A.2 bounds — the anisotropy enters only through the scalar prefactor, since $\Sigma^{1/2}u_R/\|\Sigma^{1/2}u_R\| = \nu$ is Haar regardless of Σ. Carrying the overlap-region analysis of Appendix A through with this varying prefactor is the remaining step, and the one this paper leaves open.

Remark B.6 (What the network application needs). The certification-gap theorem (Theorem 8.3) uses only the *proved* bracket: a pattern-uniform certificate must tolerate the worst-conditioned pattern, whose least singular value is at most $e^{-2H(\alpha)N/d_{\mathbb{F}}}$ by the proved achievability of Proposition B.2 and at most $e^{-(k+1)H(\alpha)N/d_{\mathbb{F}}}$ at the sharp (numerically determined) rate. So the gap is proved to be at least $e^{2H(\alpha)N/d_{\mathbb{F}}}$ and equals $e^{(k+1)H(\alpha)N/d_{\mathbb{F}}}$ at the sharp rate; the qualitative conclusion — that pattern-uniform certificates degrade exponentially in N — is unconditional.

Appendix C. Operator-norm brackets and the gate constants

This appendix proves the operator-norm brackets of Section 3 and the order-statistic and convex-hull facts used by the network estimates. Throughout, a Gaussian matrix has entry variance $1/q$ where q is the number of columns, unless a different normalization is stated.

C.1. **The order-statistic law.**

Lemma C.1 (Order statistics of squared Gaussians)**.** *Let $X_1, \dots, X_p$ be i.i.d. nonnegative integrable with continuous law and let $X_1^{\downarrow} \ge \cdots \ge X_p^{\downarrow}$ be the decreasing rearrangement. If $q/p \to \alpha \in (0,1]$ then, almost surely and in L^1,*

$$\frac{1}{p}\sum_{j=1}^{q} X_j^{\downarrow} \;\to\; \mathbb{E}[X\mathbb{1}\{X \ge \tau(\alpha)\}], \qquad \frac{1}{p}\sum_{j=p-q+1}^{p} X_j^{\downarrow} \;\to\; \mathbb{E}[X\mathbb{1}\{X \le \tau'(\alpha)\}],$$

where $\mathbb{P}(X \ge \tau(\alpha)) = \alpha$ and $\mathbb{P}(X \le \tau'(\alpha)) = \alpha$. Applied to $X_i = Z_i^2$ with $Z_i \sim N(0,1)$ these limits are $\kappa(\alpha)$ and $\check{\kappa}(\alpha)$ of (2)–(3); the deviation probability at any fixed threshold is e^{-cp} by bounded-differences concentration.

Proof. Use the variational identity $\sum_{j\le q} X_j^{\downarrow} = \min_{\lambda\ge 0}\big(q\lambda + \sum_i (X_i - \lambda)_+\big)$. The empirical mean $p^{-1}\sum_i (X_i-\lambda)_+ \to \mathbb{E}(X-\lambda)_+$ uniformly in λ on compacts by the law of large numbers and monotonicity, and $q/p \to \alpha$; the minimizer is $\lambda = \tau(\alpha)$ (the α-quantile), giving $\alpha\tau(\alpha) + \mathbb{E}(X - \tau(\alpha))_+ = \mathbb{E}[X\mathbb{1}\{X \ge \tau(\alpha)\}]$. The bottom sum is $\sum_j X_j^{\downarrow} - \sum_{j\le p-q} X_j^{\downarrow}$, the same law with $1-\alpha$. Concentration: each summand changes the statistic by a bounded amount after truncation, so McDiarmid applies with the exponential rate. □

C.2. **Operator-norm brackets.**

Proof of Proposition 3.5. Write $M_1(\alpha) = \max_{|S|=|T|=\alpha N} \|P_T A P_S\|_{\mathrm{op}} = \sup\{\langle y, Ax\rangle : \|x\| = \|y\| = 1,\ |\operatorname{supp} x| \le \alpha N,\ |\operatorname{supp} y| \le \alpha N\}$. The two index sets are αN-sparse unit spheres $\Sigma_{\mathrm{in}}, \Sigma_{\mathrm{out}}$, whose Gaussian widths are, by Lemma C.1, $w(\Sigma_{\mathrm{in}}) = \mathbb{E}\sup_{x\in\Sigma_{\mathrm{in}}}\langle g, x\rangle = \sqrt{N\kappa(\alpha)}(1+o(1))$ and likewise $\sqrt{N\kappa(\alpha)}$ for the output. Chevet's inequality bounds $\mathbb{E}M_1(\alpha) \le \operatorname{rad}(\Sigma_{\mathrm{out}})\, w(\Sigma_{\mathrm{in}})/\sqrt{N} + \operatorname{rad}(\Sigma_{\mathrm{in}})\, w(\Sigma_{\mathrm{out}})/\sqrt{N} = 2\sqrt{\kappa(\alpha)}$ with unit radii and entry variance $1/N$, and Gaussian concentration (M_1 is 1-Lipschitz in the Frobenius norm, variance $1/N$) removes the expectation, giving the upper bound $2\sqrt{\kappa(\alpha)} + o(1)$. For the lower bounds, a fixed $\alpha N \times \alpha N$ submatrix already has $\|P_T A P_S\|_{\mathrm{op}} \to 2\sqrt{\alpha}$ (Bai–Yin), and for a fixed unit x_0 on S, $Ax_0 \sim N(0, I/N)$, whose top αN coordinates have norm $\sqrt{\kappa(\alpha)} + o(1)$ by Lemma C.1, so $\max_T \|P_T A x_0\| \ge \sqrt{\kappa(\alpha)}$. □

Proof of Proposition 3.6. Upper: submultiplicativity $\|M\|_{\mathrm{op}} \le \prod_j \|F_j\|_{\mathrm{op}}$ and then relax the shared supports to independent per-factor maxima, each bounded by $2\sqrt{\kappa(\alpha)} + o(1)$ from Proposition 3.5. Lower: sequential planting. Fix a unit x_0 on S_0; at step j, x_{j-1} is a fixed vector, $A_j x_{j-1} \sim N(0, I/N)$, and choosing S_j the top αN coordinates gains $\sqrt{\kappa(\alpha)}$, with x_j its normalization, a function of $A_1, \dots, A_j$ only, so the next step is independent; this gives the greedy lower bound $\kappa(\alpha)^{k/2}\cdot(\ge \sqrt{\kappa(\alpha)})$. Boosting the first step by a fixed submatrix top singular pair ($2\sqrt{\alpha}$ by Bai–Yin) and keeping its left singular vector gives the $\max\{2\sqrt{\alpha}, \sqrt{\kappa(\alpha)}\}\kappa(\alpha)^{(k-1)/2}$ form; only one end can be boosted, since boosting fixes the adjacent output direction. □

Proof of Proposition 3.8. From Proposition 3.6, $\max_{S_\bullet} \|M\|_{\mathrm{op}} \le (2\sqrt{\kappa(\alpha)})^k$, which tends to 0 iff $2\sqrt{\kappa(\alpha)} < 1$, i.e. $\kappa(\alpha) < 1/4$; the root of $\kappa(\alpha^*) = 1/4$ is $\alpha^* = 0.042672$ numerically. The fixed-support threshold is the Fuss–Catalan edge $\alpha^{k/2}\sqrt{(k+1)^{k+1}/k^k} < 1$, i.e. $\alpha < (k^k/(k+1)^{k+1})^{1/k}$. □

C.3. **The subgaussian operator norm.**

Lemma C.2 (Subgaussian bracket)**.** *Let A be $N \times N$ with i.i.d. mean-zero, variance-$1/N$ entries of subgaussian norm $\|\sqrt{N}A_{ij}\|_{\psi_2} \le K$. Then* $\limsup_N M_1(\alpha) \le C_0K(\sqrt{\kappa(\alpha)}+\sqrt{\kappa(\alpha)}) = 2C_0K\sqrt{\kappa(\alpha)}$ *w.h.p., with C_0 absolute. The lower bracket* $\max\{2\sqrt{\alpha}, \sqrt{\kappa(\alpha)}\}$ *is unchanged and universal.*

Proof. Write $M_1(\alpha) = \sup_{x\in\Sigma_{\text{in}}, y\in\Sigma_{\text{out}}} \langle y, Ax\rangle$ and apply the subgaussian bilinear comparison (a Chevet inequality for subgaussian matrices, e.g. via a chaining/generic-chaining bound on the sparse spheres): $\mathbb{E}\sup\langle y, Ax\rangle \le C_0K(\text{rad}(\Sigma_{\text{out}})w(\Sigma_{\text{in}}) + \text{rad}(\Sigma_{\text{in}})w(\Sigma_{\text{out}}))/\sqrt{N}$, with the same widths $\sqrt{N\kappa(\alpha)}$. Bousquet-type concentration for suprema of subgaussian processes removes the expectation. The lower bound is a fixed submatrix and a fixed direction, both of which only use the variance. □

C.4. **The convex-hull reduction.**

Lemma C.3 (Gates in the hull)**.** *Let W be a fixed matrix and $\mathcal{D}_\pi$ the set of diagonal matrices with entries in $[0,1]$ and support pattern π. Then* $\sup_{D\in\mathcal{D}_\pi} \|DWx\| = \max_{D\in\{0,1\}^\pi} \|DWx\|$ *for every x, and* $\inf_{D\in\mathcal{D}_\pi,\, D\succeq D_0} \sigma_{\min}(DW) \ge \sigma_{\min}(D_0W)$ *for $D_0 \in \{0,1\}^\pi$. Consequently every operator-norm upper bound and least-singular-value lower bound proved for $0/1$ gates holds for any $[0,1]$-slope activation with the same support pattern.*

Proof. $D \mapsto \|DWx\|^2 = \sum_i D_{ii}^2 (Wx)_i^2$ is convex in the diagonal entries, hence maximized at a vertex $D \in \{0,1\}^\pi$; and $D \mapsto \sigma_{\min}(DW)$ is monotone on the order interval (increasing a nonnegative diagonal entry cannot decrease any singular value, by the min–max characterization applied to W^*D^2W), so the least singular value over $D \succeq D_0$ is at least $\sigma_{\min}(D_0W)$. The forward-pass and gradient-lower-bound estimates use exactly these two extremal facts. □

Appendix D. Network lemmas

This appendix completes the proofs of the uniform gain, gradient lower bound, and stability estimates of Section 5. The random-matrix inputs are Lemmas C.1 and C.3 and Proposition 3.5 of Appendix C; the network structure is that of Section 4.

D.1. **Uniform two-sided gain.**

Proof of Theorem 5.1. Fix a unit $x \in V$. The vector Wx has i.i.d. $N(0, 1/q)$ coordinates, so $q\|Wx\|^2$ is a chi-square with p degrees of freedom and, by Lemma C.1 applied to $Z_i^2 = (q^{1/2}(Wx)_i)^2$, the top and bottom αp squared coordinates satisfy

$$\frac{1}{q}\sum_{i\in T_{\text{top}}} (Wx)_i^2 \to \tfrac{p}{q}\kappa(\alpha), \qquad \frac{1}{q}\sum_{i\in T_{\text{bot}}} (Wx)_i^2 \to \tfrac{p}{q}\check{\kappa}(\alpha),$$

with deviation probability e^{-cp} at any fixed threshold. For an arbitrary T of size αp, $\sum_{i\in T}(Wx)_i^2$ lies between the bottom and top sums, giving the two-sided bracket for that x and that T, with the extremes attained by $T = T_{\text{top}}, T_{\text{bot}}$. To make this uniform over $x \in V$, take an ε-net $\mathcal{N}$ of the unit sphere of V, $|\mathcal{N}| \le (3/\varepsilon)^d$; apply the bound at each net point with threshold slack ε, and union bound: the failure probability is $(3/\varepsilon)^d e^{-cp} \to 0$ since $d\log(1/\varepsilon) = o(p)$. For a general x, pick $x' \in \mathcal{N}$ with $\|x - x'\| \le \varepsilon$; then $\|P_TWx\|$ and $\|P_TWx'\|$ differ by at most $\|W\|_{\text{op}}\varepsilon \le C\sqrt{p/q}\,\varepsilon$, and the top/bottom-$\alpha$ selections for x and x' differ on at most $O(\varepsilon p)$ coordinates, whose contribution is $O(\sqrt{\varepsilon})$; choosing $\varepsilon = 1/p$ makes both corrections $o(1)$. This gives the uniform two-sided bracket and the sharpness. □

D.2. **Gradient lower bound.**

Proof of Proposition 5.2, details. For a fixed input x, the symmetric-weight decoupling of [3, Lemma 2.1] gives that $\nabla F(x)^\top = w^\top D_L(x)W_L \cdots D_1(x)W_1$ has the same law as $w^\top \Sigma_L W_L \cdots \Sigma_1 W_1$, where $\Sigma_\ell = \mathrm{diag}(\xi_{\ell,1}, \dots, \xi_{\ell,n_\ell})$ has entries $\xi_{\ell,i}$ i.i.d. with the law of $\sigma'_\ell(Z)$ (Z the preactivation), independent of the W's. Set $v_0 = w$ and $v_\ell^\top = v_{\ell-1}^\top \Sigma_\ell W_\ell$; then $\mathbb{E}[\|v_\ell\|^2 \mid v_{\ell-1}] = \mathbb{E}[\Sigma_\ell]^2$-weighted $\|v_{\ell-1}\|^2$ times the layer variance, and the pointwise nondegeneracy $\sigma'_\ell(r)^2 + \sigma'_\ell(-r)^2 \geq 2c^2$ combined with the signal-preserving variance gives $\mathbb{E}[\|v_\ell\|^2 \mid v_{\ell-1}] \geq c_0^2 \|v_{\ell-1}\|^2$ for an absolute $c_0 > 0$ (for ReLU, $\mathbb{E}\xi_{\ell,i}^2 = 1/2$ and the variance-2 normalization make the per-layer factor $\geq 1 - o(1)$). Iterating, $\mathbb{E}\|\nabla F(x)\|^2 \geq c_g^2 \|w\|^2$ with $c_g = c_0^L$; concentration of the quadratic form over the $\Theta(\prod_\ell n_\ell)$ Gaussian entries (Hanson–Wright) gives $\|\nabla F(x)\| \geq \frac{1}{2} c_g \|w\|$ with failure probability $e^{-c' \min_\ell n_\ell}$. Uniformity over x is by the ε-net argument as in Theorem 5.1, extended to all x by Theorem 5.3. The nondegeneracy holds for any $[0,1]$-slope σ with $\mathbb{E}\sigma'(Z)^2 > 0$ by Lemma C.3 (the derivative gate lies in the hull of 0/1 gates and the extremal least-singular bound transfers); biases shift the gate density, keeping it in $[c_0, c_1]$ by Assumption 4.2, which only changes c_0. □

D.3. **Stability.**

Lemma D.1 (Few gates flip)**.** *Under the hypotheses of Proposition 5.2, w.h.p. for all inputs x on the sphere and x' with $\|x' - x\| \leq r$, the number of neurons at layer ℓ whose gate differs between x and x' is at most $\delta_\ell n_\ell$ with $\delta_\ell = Cr\sqrt{\bar{n}}/\|x\|$.*

Proof. A neuron at layer ℓ flips only if its preactivation crosses a breakpoint of σ'_ℓ, i.e. if $z_{\ell,i}(x)$ lies within $\|z_\ell(x') - z_\ell(x)\|_\infty$ of a breakpoint. The layer maps are $O(1)$-Lipschitz (upper bracket of Theorem 5.1), so $\|z_\ell(x') - z_\ell(x)\|_\infty \leq \|z_\ell(x') - z_\ell(x)\|_2 \leq Cr$ after the forward Lipschitz product. The preactivations $z_{\ell,i}$ are Gaussian of scale $\varsigma_\ell \asymp \|h_{\ell-1}\|/\sqrt{n_{\ell-1}} \asymp \|x\|/\sqrt{n_{\ell-1}}$, so by anticoncentration $\mathbb{P}(|z_{\ell,i} - \text{breakpoint}| \leq \tau) \leq C\tau/\varsigma_\ell$ with $\tau = Cr$; the expected number of flipped neurons is $n_\ell \cdot Cr\sqrt{n_{\ell-1}}/\|x\| \leq Cr\sqrt{\bar{n}}/\|x\| \cdot n_\ell$, and a Bernstein bound gives the same w.h.p. uniformly over x, x' by a net (the flip count is a Lipschitz function of x, x' after smoothing the indicator). □

Proof of Theorem 5.3, details. By the telescoping identity in the sketch, $\nabla F(x')^\top - \nabla F(x)^\top$ is a sum of L terms, the ℓ-th being $w^\top D_L(x')W_L \cdots (D_\ell(x') - D_\ell(x))W_\ell \cdots D_1(x)W_1$. The gate difference $D_\ell(x') - D_\ell(x)$ has entries in $[-1, 1]$ supported on the flipped neurons, of number $\leq \delta_\ell n_\ell$ with $\delta_\ell = Cr\sqrt{\bar{n}}/\|x\|$ by Lemma D.1, so $(D_\ell(x') - D_\ell(x))W_\ell$ restricted to that support has operator norm at most $2\sqrt{\kappa(\delta_\ell)} + o(1)$ by Proposition 3.5 (a Gaussian block on a density-δ_ℓ row pattern), and $\kappa(\delta) \leq C\delta \log(1/\delta)$ as $\delta \to 0$ (the top-δ Gaussian square-mass). The remaining factors are $O(1)$ in operator norm by Theorem 5.1, and the leading $w^\top D_L(\cdots)$ contributes $\|w\|$. Hence the ℓ-th term is at most $C\|w\|\sqrt{\delta_\ell \log(1/\delta_\ell)}$, and summing over ℓ (all $\delta_\ell \leq \delta := Cr\sqrt{\bar{n}}/\|x\|$) gives $\|\nabla F(x') - \nabla F(x)\| \leq C(L)\|w\|\sqrt{\delta \log(1/\delta)}$, which is the stated bound. Uniformity over x, x' is again by a net and the $O(1)$-Lipschitz dependence of every factor on the inputs. □

D.4. **Combining into (A3).** For the master theorem we need $\|\nabla F(x') - \nabla F(x)\| \leq C_g/3$ on the attack ball of radius $2a/C_g$. With $a = C\|x\|\sqrt{(d \log \bar{n})/n_L}$ and $C_g = c_g\|w\|$, the radius is $r = 2a/C_g \asymp \|x\|\sqrt{(d \log \bar{n})/n_L}/c_g$, so $\delta = Cr\sqrt{\bar{n}}/\|x\| \asymp \sqrt{(d \log \bar{n})\bar{n}/n_L}/c_g$, and $\|\nabla F(x') - \nabla F(x)\| \leq C(L)\|w\|\sqrt{\delta \log(1/\delta)} = o(\|w\|) = o(C_g)$ whenever $\bar{n}\, d \log \bar{n} = o(n_L\, c_g^2)$; when the widths are comparable the directional (attack-line) form of Theorem 5.3 and the constant-2 Taylor engine of [1, Thm 4.6] require only $\|\nabla F(x') - \nabla F(x)\| = o(\|\nabla F\|)$ along the segment, which holds under $d \log n = o(n)$ (Remark 7.3).

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